

Final Project

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Data Preparation

```
heart_data <- read.csv("heart_2020_cleaned.csv", header = TRUE)
heart_data$HeartDisease <- factor(heart_data$HeartDisease, levels =
c("No", "Yes"), labels= c("No", "Yes"))
heart_data$HeartDisease <- relevel(heart_data$HeartDisease, ref = "Yes") #
"Yes" group the positive reference group
cat_vars <- c("Smoking", "AlcoholDrinking", "Stroke",
"DiffWalking", "Sex", "AgeCategory", "Race", "Diabetic", "PhysicalActivity",
"GenHealth", "Asthma", "KidneyDisease", "SkinCancer")
num_vars <- c("BMI", "PhysicalHealth", "MentalHealth",
"SleepTime", "AgeCategory.num", "GenHealth.num")
#Potential New Variables
genhealth_levels<- c("Poor", "Fair", "Good", "Very good", "Excellent")
age_levels<- c("18-24", "25-29", "30-34", "35-39", "40-44", "45-49", "50-54", "55-
59", "60-64", "65-69", "70-74", "75-79", "80 or older")
heart_data <- heart_data %>%
  mutate(GenHealth.num = as.numeric(factor(GenHealth, levels =
genhealth_levels, ordered = TRUE))),
  AgeCategory.num = as.numeric(factor(AgeCategory, levels = age_levels,
ordered = TRUE)))
heart_data <- heart_data %>% mutate(across(all_of(cat_vars), as.factor))
heart_data <- heart_data %>% mutate(across(all_of(num_vars), as.numeric))
# Make new dataset
factor_cols <- sapply(heart_data, is.factor) # Identify factor columns
level_counts <- sapply(heart_data[, factor_cols, drop = FALSE], nlevels) #
Count Levels
ordered_factor_col_names <- names(level_counts)[order(level_counts)] # Order
factor columns (by # of levels)
non_factor_col_names <- names(heart_data)[!factor_cols] # Get non-factor
column names
new_col_order <- c(ordered_factor_col_names, non_factor_col_names) # Combine
data_sorted <- heart_data[, new_col_order] # Reorder the data frame columns
write.csv(data_sorted, "heart2020_data_sorted.csv", row.names = FALSE)
```

Data Setup

```
heart_data = read.csv("heart2020_data_sorted.csv", header=TRUE)
heart_data <- heart_data %>% mutate(across(all_of(cat_vars), as.factor))
heart_data <- heart_data %>% mutate(across(all_of(num_vars), as.numeric))
```

Overall Dataset

```
# Table of percent and number counts
heart = heart_data %>% group_by(HeartDisease) %>%
```

```

  summarise(n = n()) %>%
  mutate(percent=round(n/sum(n),4)*100)
heart

## # A tibble: 2 × 3
##   HeartDisease      n percent
##   <chr>          <int>   <dbl>
## 1 No            292422    91.4
## 2 Yes           27373     8.56

```

Data Split

```

library(caret)
set.seed(4)
trainIndex <- createDataPartition(heart_data$HeartDisease, p = 0.8, list =
FALSE) # to get balance of yes/no
trainHeart <- heart_data[trainIndex, ]
testHeart <- heart_data[-trainIndex, ]
write.csv(trainHeart, "trainHeart.csv", row.names = FALSE)
write.csv(testHeart, "testHeart.csv", row.names = FALSE)

trainHeart <- read.csv("trainHeart.csv")
testHeart <- read.csv("testHeart.csv")
trainHeart$HeartDisease <- factor(trainHeart$HeartDisease, levels =
c("No", "Yes"))
testHeart$HeartDisease <- factor(testHeart$HeartDisease, levels =
c("No", "Yes"))
table(trainHeart$HeartDisease)

##
##      No      Yes
## 233938  21899

table(testHeart$HeartDisease)

##
##      No      Yes
## 58484   5474

```

EDA: Training Dataset

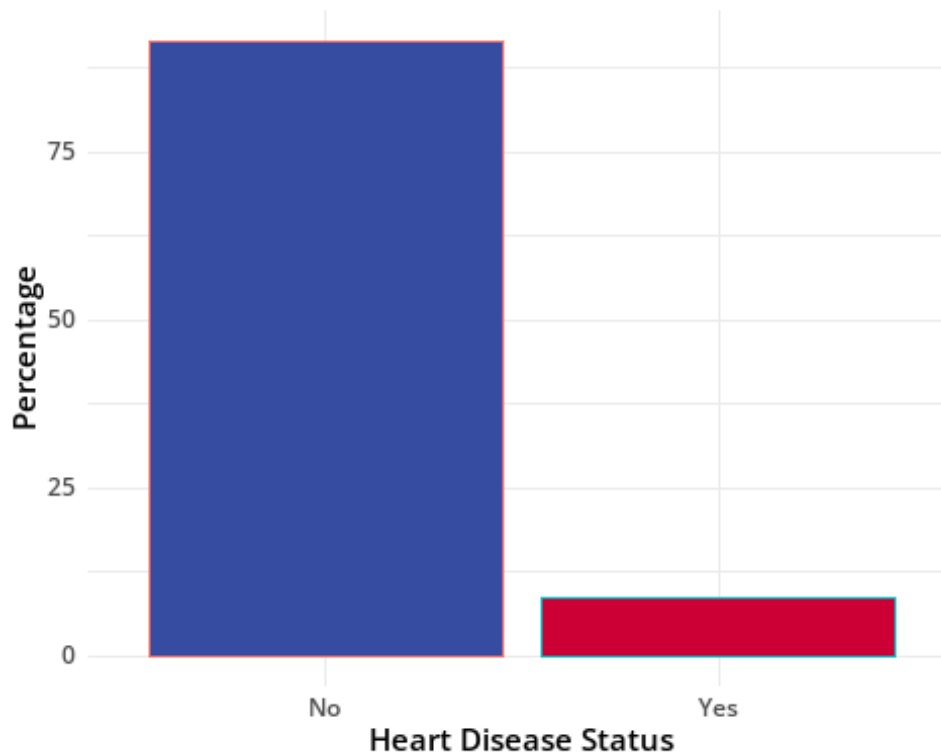
```

# Table of percent and number counts
heartTrain = trainHeart %>% group_by(HeartDisease) %>%
  summarise(n = n()) %>%
  mutate(percent=round(n/sum(n),4)*100)
heartTrain

## # A tibble: 2 × 3
##   HeartDisease      n percent
##   <fct>          <int>   <dbl>
## 1 No            233938    91.4
## 2 Yes           21899     8.56

```

```
# Percent of disease (yes/no)
ggplot(heartTrain,aes(x=HeartDisease,y=percent,colour=HeartDisease))+geom_bar(
  aes(fill=HeartDisease),show.legend=F,stat="identity")+ylab("Percentage") +
  scale_fill_manual(values = c("Yes" = "#CC0035", "No" = "#354CA1")) +
  theme_minimal(base_family = "Open Sans") + xlab("Heart Disease Status")
```



```
trainHeart$HeartDisease.num =
as.numeric(ifelse(trainHeart$HeartDisease=="Yes",1,0))
numeric_data <- trainHeart %>% select(where(is.numeric))
numeric_data %>% filter(HeartDisease.num == "0") %>% summary()
```

##	BMI	PhysicalHealth	MentalHealth	SleepTime
##	Min. :12.02	Min. : 0.000	Min. : 0.000	Min. : 1.000
##	1st Qu.:23.89	1st Qu.: 0.000	1st Qu.: 0.000	1st Qu.: 6.000
##	Median :27.26	Median : 0.000	Median : 0.000	Median : 7.000
##	Mean :28.22	Mean : 2.954	Mean : 3.822	Mean : 7.092
##	3rd Qu.:31.32	3rd Qu.: 1.000	3rd Qu.: 3.000	3rd Qu.: 8.000
##	Max. :94.66	Max. :30.000	Max. :30.000	Max. :24.000
##	GenHealth.num	AgeCategory.num	HeartDisease.num	
##	Min. :1.000	Min. : 1.000	Min. : 0	
##	1st Qu.:3.000	1st Qu.: 4.000	1st Qu.: 0	
##	Median :4.000	Median : 8.000	Median : 0	
##	Mean :3.673	Mean : 7.259	Mean : 0	
##	3rd Qu.:4.000	3rd Qu.:10.000	3rd Qu.: 0	
##	Max. :5.000	Max. :13.000	Max. : 0	

```
numeric_data %>% filter(HeartDisease.num == "1") %>% summary()
```

```
##      BMI      PhysicalHealth      MentalHealth      SleepTime
## Min.   :12.21   Min.    : 0.000   Min.    : 0.000   Min.    : 1.000
## 1st Qu.:25.06   1st Qu.: 0.000   1st Qu.: 0.000   1st Qu.: 6.000
## Median :28.32   Median : 0.000   Median : 0.000   Median : 7.000
## Mean   :29.38   Mean    : 7.839   Mean    : 4.665   Mean    : 7.146
## 3rd Qu.:32.61   3rd Qu.:15.000   3rd Qu.: 4.000   3rd Qu.: 8.000
## Max.   :83.33   Max.    :30.000   Max.    :30.000   Max.    :24.000
## GenHealth.num AgeCategory.num HeartDisease.num
## Min.   :1.000   Min.    : 1.00   Min.    :1
## 1st Qu.:2.000   1st Qu.: 9.00   1st Qu.:1
## Median :3.000   Median :11.00   Median :1
## Mean   :2.762   Mean    :10.23   Mean    :1
## 3rd Qu.:4.000   3rd Qu.:12.00   3rd Qu.:1
## Max.   :5.000   Max.    :13.00   Max.    :1
```

Correlation Plot and Table (Numeric Only)

```
colors <- colorRampPalette(c("#354CA1", "white", "#CC0035"))(200)
```

```
# Correlation Table
```

```
numeric_data %>%
```

```
  cor() %>%
```

```
  round(2)
```

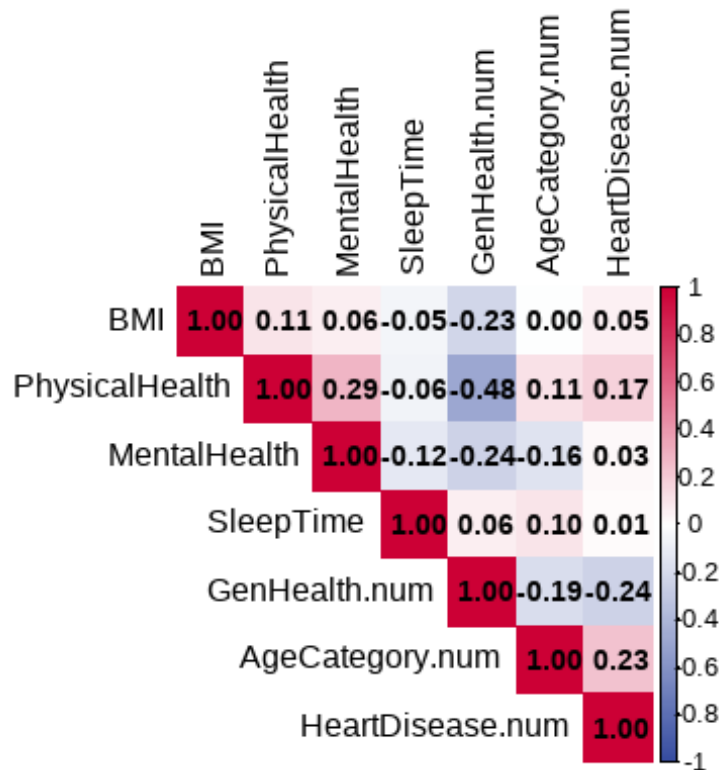
```
##      BMI      PhysicalHealth      MentalHealth      SleepTime      GenHealth.num
## BMI      1.00      0.11      0.06      -0.05      -0.23
## PhysicalHealth 0.11      1.00      0.29      -0.06      -0.48
## MentalHealth  0.06      0.29      1.00      -0.12      -0.24
## SleepTime     -0.05     -0.06     -0.12      1.00      0.06
## GenHealth.num -0.23     -0.48     -0.24      0.06      1.00
## AgeCategory.num 0.00      0.11     -0.16      0.10     -0.19
## HeartDisease.num 0.05      0.17      0.03      0.01     -0.24
##
##      AgeCategory.num      HeartDisease.num
## BMI      0.00      0.05
## PhysicalHealth 0.11      0.17
## MentalHealth -0.16      0.03
## SleepTime    0.10      0.01
## GenHealth.num -0.19     -0.24
## AgeCategory.num 1.00      0.23
## HeartDisease.num 0.23      1.00
```

```
# Correlation Plot
```

```
numeric_data %>%
```

```
  cor() %>%
```

```
  corplot(method = "color", type = "upper", col = colors, tl.col = "black",
addCoef.col = "black", number.cex = 0.9)
```



Scatterplot Matrix of Numeric Variables

```
numeric_data$HeartDisease = trainHeart$HeartDisease
numeric_data$HeartDisease <- factor(numeric_data$HeartDisease, levels =
c("No", "Yes"))
numeric_data$HeartDisease <- relevel(numeric_data$HeartDisease, ref = "Yes")
numeric_datapair = numeric_data %>% select(-HeartDisease.num)

smu_colors <- c("No" = "#354CA1", "Yes" = "#CC0035")
p <- ggpairs(numeric_datapair, mapping = aes(colour = HeartDisease), lower =
list(continuous = wrap("points", alpha = 0.3, size = 0.1)))

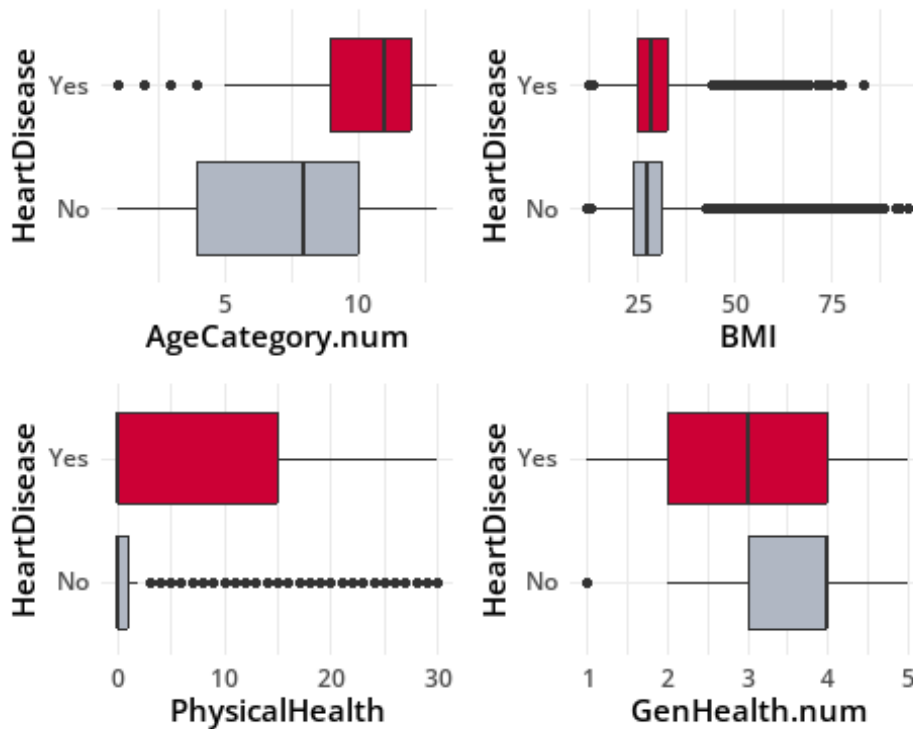
p + scale_color_manual(values = smu_colors) + scale_fill_manual(values =
smu_colors)
```

Box Plots of Numeric Predictors

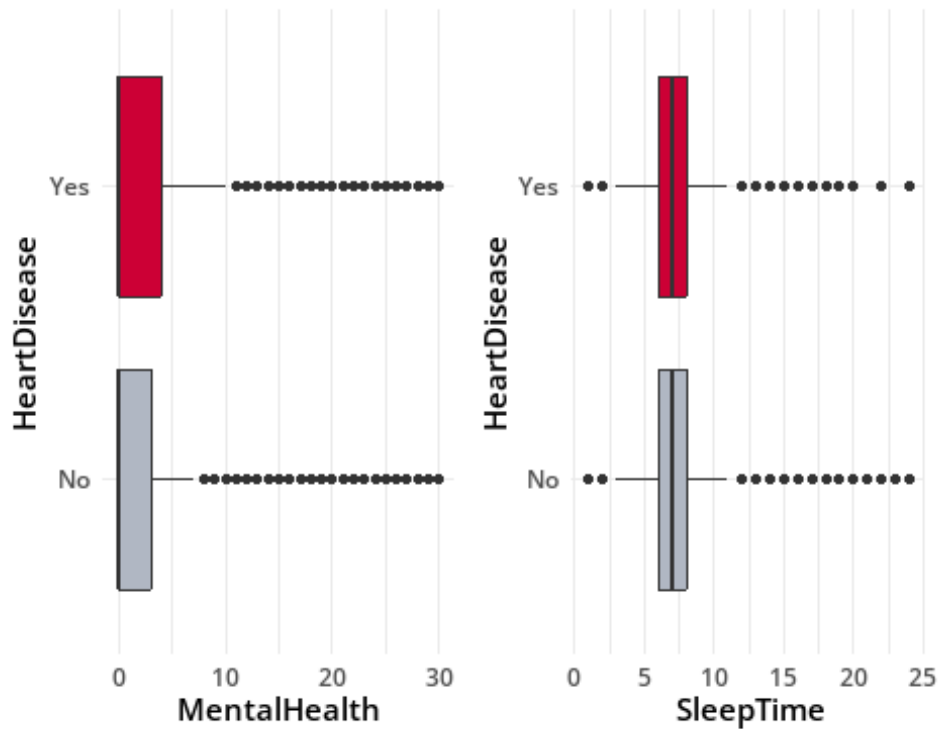
```
smu2 <- c("Yes" = "#CC0035", "No" = "#B0B7C3")
a = trainHeart %>% ggplot(aes(x=BMI, y=HeartDisease, fill = HeartDisease)) +
geom_boxplot(show.legend = FALSE) + scale_fill_manual(values = smu2) +
theme_minimal(base_family = "Open Sans")
b = trainHeart %>% ggplot(aes(x=SleepTime, y=HeartDisease, fill =
HeartDisease)) + geom_boxplot(show.legend = FALSE) + scale_fill_manual(values
= smu2) + theme_minimal(base_family = "Open Sans")
c = trainHeart %>% ggplot(aes(x=PhysicalHealth, y=HeartDisease, fill =
HeartDisease)) + geom_boxplot(show.legend = FALSE) + scale_fill_manual(values
= smu2) + theme_minimal(base_family = "Open Sans")
```

```
d = trainHeart %>% ggplot(aes(x=MentalHealth, y=HeartDisease, fill =
HeartDisease)) + geom_boxplot(show.legend = FALSE)+ scale_fill_manual(values
= smu2)+ theme_minimal(base_family = "Open Sans")
e = trainHeart %>% ggplot(aes(x=AgeCategory.num, y=HeartDisease, fill =
HeartDisease)) + geom_boxplot(show.legend = FALSE)+ scale_fill_manual(values
= smu2)+theme_minimal(base_family = "Open Sans")
f = trainHeart %>% ggplot(aes(x=GenHealth.num, y=HeartDisease, fill =
HeartDisease)) + geom_boxplot(show.legend = FALSE)+ scale_fill_manual(values
= smu2)+theme_minimal(base_family = "Open Sans")

# More visual differences
(e|a)/(c|f)
```



```
# Less visual differences
(d|b)
```



Variables with more than two levels: EDA

```
# g1: AgeCategory
g1<-trainHeart %>%
  group_by(AgeCategory,HeartDisease) %>%
  summarise(cnt=n()) %>%
  mutate(perc=round(cnt/sum(cnt),4))%>%
  arrange(desc(perc))
g1
```

```
## # A tibble: 26 × 4
## # Groups:   AgeCategory [13]
##   AgeCategory HeartDisease cnt  perc
##   <chr>         <fct>    <int> <dbl>
## 1 18-24         No      16665 0.994
## 2 25-29         No      13504 0.992
## 3 30-34         No      14858 0.988
## 4 35-39         No      16293 0.986
## 5 40-44         No      16451 0.977
## 6 45-49         No      16789 0.966
## 7 50-54         No      19074 0.945
## 8 55-59         No      22024 0.926
## 9 60-64         No      24290 0.901
## 10 65-69        No      24181 0.880
## # 16 more rows
```

```
# g2: GenHealth
g2<-trainHeart %>%
  group_by(GenHealth,HeartDisease) %>%
  summarise(cnt=n()) %>%
  mutate(perc=round(cnt/sum(cnt),4))%>%
  arrange(desc(perc))
```

g2

```
## # A tibble: 10 × 4
## # Groups:   GenHealth [5]
##   GenHealth HeartDisease    cnt  perc
##   <chr>      <fct>      <int> <dbl>
## 1 Excellent No          52294 0.978
## 2 Very good No          86841 0.953
## 3 Good      No          66793 0.898
## 4 Fair      No          22052 0.795
## 5 Poor      No           5958 0.658
## 6 Poor      Yes           3095 0.342
## 7 Fair      Yes           5702 0.205
## 8 Good      Yes           7618 0.102
## 9 Very good Yes          4281 0.047
## 10 Excellent Yes          1203 0.0225
```

```
# g3: Diabetic
```

```
g3<-trainHeart %>%
  group_by(Diabetic,HeartDisease) %>%
  summarise(cnt=n()) %>%
  mutate(perc=round(cnt/sum(cnt),4))%>%
  arrange(desc(perc))
```

g3

```
## # A tibble: 8 × 4
## # Groups:   Diabetic [4]
##   Diabetic      HeartDisease    cnt  perc
##   <chr>      <fct>      <int> <dbl>
## 1 Yes (during pregnancy) No          1974 0.954
## 2 No          No          201753 0.935
## 3 No, borderline diabetes No           4735 0.884
## 4 Yes          No          25476 0.782
## 5 Yes          Yes           7116 0.218
## 6 No, borderline diabetes Yes           623 0.116
## 7 No          Yes          14064 0.0652
## 8 Yes (during pregnancy) Yes           96 0.0464
```

```
# g4: Race
```

```
g4 <- trainHeart %>%
  group_by(Race, HeartDisease) %>%
  summarise(cnt = n()) %>%
  mutate(perc = round(cnt / sum(cnt), 4)) %>%
  arrange(desc(perc))
```

g4


```
## # A tibble: 12 × 4
## # Groups:   Race [6]
##   Race                HeartDisease    cnt    perc
##   <chr>                <fct>      <int>  <dbl>
## 1 Asian                No          6229  0.968
## 2 Hispanic              No         20836  0.947
## 3 Black                 No         16976  0.924
## 4 Other                 No          8044  0.918
## 5 White                 No        178135  0.908
## 6 American Indian/Alaskan Native No          3718  0.894
## 7 American Indian/Alaskan Native Yes           440  0.106
## 8 White                 Yes         17993  0.0917
## 9 Other                 Yes           715  0.0816
## 10 Black                Yes          1388  0.0756
## 11 Hispanic              Yes          1156  0.0526
## 12 Asian                 Yes           207  0.0322
```

Proportion plots grouped by HeartDisease

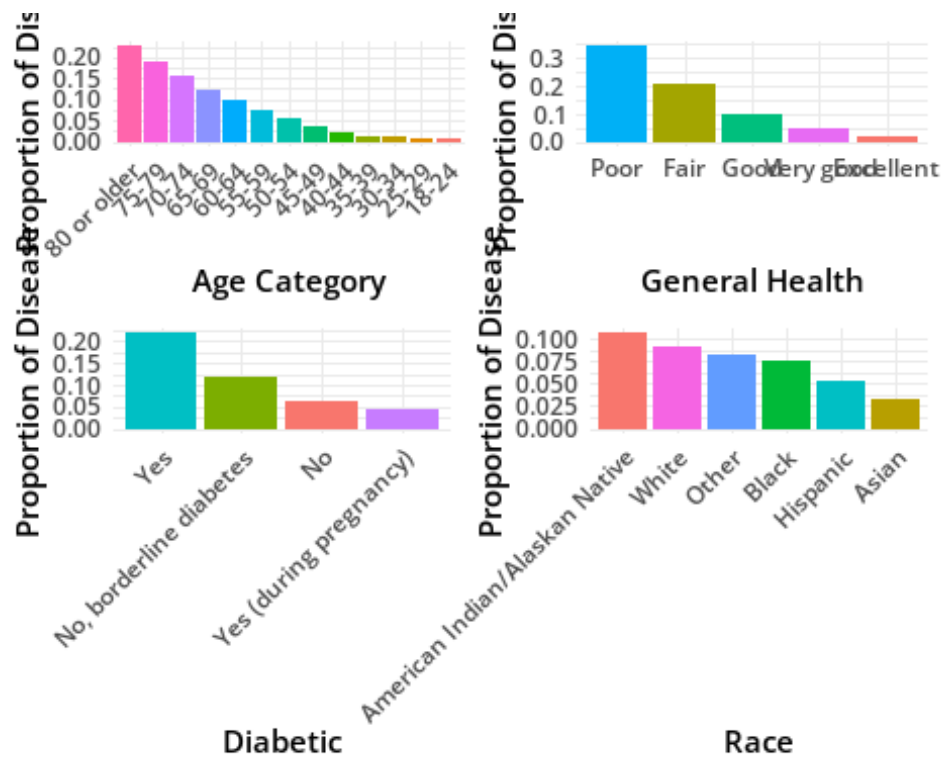
```
g1g <- ggplot(g1[c(14:26),], aes(x = reorder(AgeCategory, -perc), y = perc,
fill = AgeCategory)) +
  geom_bar(stat = "identity", show.legend = FALSE) + ylab("Proportion of
Disease") + xlab("Age Category") +
  theme_minimal(base_family = "Open Sans") + theme(axis.text.x =
element_text(angle = 45, hjust = 1))

g2g <- ggplot(g2[c(6:10),], aes(x = reorder(GenHealth, -perc), y = perc, fill
= GenHealth)) +
  geom_bar(stat = "identity", show.legend = FALSE) + ylab("Proportion of
Disease") + xlab("General Health") + theme_minimal(base_family = "Open Sans")

g3g <- ggplot(g3[c(5:8),], aes(x = reorder(Diabetic, -perc), y = perc, fill =
Diabetic)) +
  geom_bar(stat = "identity", show.legend = FALSE) + ylab("Proportion of
Disease") + xlab("Diabetic") +
  theme_minimal(base_family = "Open Sans") + theme(axis.text.x =
element_text(angle = 45, hjust = 1))

g4g <- ggplot(g4[c(7:12),], aes(x = reorder(Race, -perc), y = perc, fill =
Race)) +
  geom_bar(stat = "identity", show.legend = FALSE) + ylab("Proportion of
Disease") + xlab("Race") +
  theme_minimal(base_family = "Open Sans") + theme(axis.text.x =
element_text(angle = 45, hjust = 1))

(g1g | g2g) / (g3g | g4g)
```



Binary Categorical Predictors EDA

```
# b1: Smoking
b1<-trainHeart %>%
  group_by(Smoking,HeartDisease) %>%
  summarise(cnt=n()) %>%
  mutate(perc=round(cnt/sum(cnt),4))%>%
  arrange(desc(perc))
b1

## # A tibble: 4 × 4
## # Groups:   Smoking [2]
##   Smoking HeartDisease    cnt  perc
##   <chr>    <fct>    <int> <dbl>
## 1 No      No      141246 0.939
## 2 Yes     No      92692 0.879
## 3 Yes     Yes     12759 0.121
## 4 No      Yes      9140 0.0608

# b2:AlcoholDrinking
b2<-trainHeart %>%
  group_by(AlcoholDrinking,HeartDisease) %>%
  summarise(cnt=n()) %>%
  mutate(perc=round(cnt/sum(cnt),4))%>%
  arrange(desc(perc))
b2
```

```
## # A tibble: 4 × 4
## # Groups:   AlcoholDrinking [2]
##   AlcoholDrinking HeartDisease    cnt    perc
##   <chr>          <fct>      <int>  <dbl>
## 1 Yes           No          16503 0.947
## 2 No            No          217435 0.912
## 3 No            Yes           20970 0.088
## 4 Yes           Yes            929 0.0533
```

b3: Stroke

```
b3 <- trainHeart %>%
  group_by(Stroke, HeartDisease) %>%
  summarise(cnt = n()) %>%
  mutate(perc = round(cnt / sum(cnt), 4)) %>%
  arrange(desc(perc))
b3
```

```
## # A tibble: 4 × 4
## # Groups:   Stroke [2]
##   Stroke HeartDisease    cnt    perc
##   <chr>  <fct>      <int>  <dbl>
## 1 No     No          227788 0.925
## 2 Yes    No           6150 0.636
## 3 Yes    Yes           3523 0.364
## 4 No     Yes          18376 0.0746
```

b4: DiffWalking

```
b4 <- trainHeart %>%
  group_by(DiffWalking, HeartDisease) %>%
  summarise(cnt = n()) %>%
  mutate(perc = round(cnt / sum(cnt), 4)) %>%
  arrange(desc(perc))
b4
```

```
## # A tibble: 4 × 4
## # Groups:   DiffWalking [2]
##   DiffWalking HeartDisease    cnt    perc
##   <chr>        <fct>      <int>  <dbl>
## 1 No          No          206433 0.937
## 2 Yes          No           27505 0.774
## 3 Yes          Yes            8006 0.226
## 4 No          Yes           13893 0.0631
```

b5: Sex

```
b5 <- trainHeart %>%
  group_by(Sex, HeartDisease) %>%
  summarise(cnt = n()) %>%
  mutate(perc = round(cnt / sum(cnt), 4)) %>%
  arrange(desc(perc))
b5
```

```
## # A tibble: 4 × 4
## # Groups:   Sex [2]
##   Sex    HeartDisease    cnt    perc
##   <chr>   <fct>         <int>  <dbl>
## 1 Female No           125326 0.933
## 2 Male   No           108612 0.894
## 3 Male   Yes            12889 0.106
## 4 Female Yes            9010 0.0671

# b6: PhysicalActivity
b6 <- trainHeart %>%
  group_by(PhysicalActivity, HeartDisease) %>%
  summarise(cnt = n()) %>%
  mutate(perc = round(cnt / sum(cnt), 4)) %>%
  arrange(desc(perc))
b6

## # A tibble: 4 × 4
## # Groups:   PhysicalActivity [2]
##   PhysicalActivity HeartDisease    cnt    perc
##   <chr>            <fct>         <int>  <dbl>
## 1 Yes              No           184380 0.930
## 2 No               No           49558 0.862
## 3 No               Yes            7935 0.138
## 4 Yes              Yes           13964 0.0704

# b7: Asthma
b7 <- trainHeart %>%
  group_by(Asthma, HeartDisease) %>%
  summarise(cnt = n()) %>%
  mutate(perc = round(cnt / sum(cnt), 4)) %>%
  arrange(desc(perc))
b7

## # A tibble: 4 × 4
## # Groups:   Asthma [2]
##   Asthma HeartDisease    cnt    perc
##   <chr>   <fct>         <int>  <dbl>
## 1 No     No           203558 0.919
## 2 Yes    No           30380 0.886
## 3 Yes    Yes            3921 0.114
## 4 No     Yes           17978 0.0812

# b8: KidneyDisease
b8 <- trainHeart %>%
  group_by(KidneyDisease, HeartDisease) %>%
  summarise(cnt = n()) %>%
  mutate(perc = round(cnt / sum(cnt), 4)) %>%
  arrange(desc(perc))
b8
```

```
## # A tibble: 4 × 4
## # Groups:   KidneyDisease [2]
##   KidneyDisease HeartDisease    cnt  perc
##   <chr>         <fct>      <int> <dbl>
## 1 No           No        227313 0.922
## 2 Yes          No         6625 0.705
## 3 Yes          Yes         2777 0.295
## 4 No           Yes        19122 0.0776
```

```
# b9: SkinCancer
```

```
b9 <- trainHeart %>%
  group_by(SkinCancer, HeartDisease) %>%
  summarise(cnt = n()) %>%
  mutate(perc = round(cnt / sum(cnt), 4)) %>%
  arrange(desc(perc))
b9
```

```
## # A tibble: 4 × 4
## # Groups:   SkinCancer [2]
##   SkinCancer HeartDisease    cnt  perc
##   <chr>         <fct>      <int> <dbl>
## 1 No           No        214127 0.923
## 2 Yes          No         19811 0.833
## 3 Yes          Yes         3964 0.167
## 4 No           Yes        17935 0.0773
```

Proportion plots grouped by HeartDisease

```
b1b <- ggplot(b1[c(3:4),], aes(x = reorder(Smoking, -perc), y = perc, fill =
Smoking)) +
  geom_bar(stat = "identity", show.legend = FALSE) + scale_fill_manual(values
= smu2) +
  labs(x = "Smoking Status", y = "Proportion with Heart Disease") +
  theme_minimal(base_family = "Open Sans")
```

```
b2b <- ggplot(b2[c(3:4),], aes(x = reorder(AlcoholDrinking, -perc), y = perc,
fill = AlcoholDrinking)) + geom_bar(stat = "identity", show.legend = FALSE) +
scale_fill_manual(values = smu2) + labs(x = "Alcohol Drinking", y =
"Proportion with Heart Disease") + theme_minimal(base_family = "Open Sans")
```

```
b3b <- ggplot(b3[c(3:4),], aes(x = reorder(Stroke, -perc), y = perc, fill =
Stroke)) + geom_bar(stat = "identity", show.legend = FALSE) +
scale_fill_manual(values = smu2) + labs(x = "Stroke", y = "Proportion with
Heart Disease") + theme_minimal(base_family = "Open Sans")
```

```
b4b <- ggplot(b4[c(3:4),], aes(x = reorder(DiffWalking, -perc), y = perc,
fill = DiffWalking)) + geom_bar(stat = "identity", show.legend = FALSE) +
scale_fill_manual(values = smu2) + labs(x = "Difficulty Walking", y =
"Proportion with Heart Disease") + theme_minimal(base_family = "Open Sans")
```

```

b5b <- ggplot(b5[c(3:4),], aes(x = reorder(Sex, -perc), y = perc, fill =
Sex)) + geom_bar(stat = "identity", show.legend = FALSE) + labs(x = "Sex", y
= "Proportion with Heart Disease") + theme_minimal(base_family = "Open Sans")

b6b <- ggplot(b6[c(3:4),], aes(x = reorder(PhysicalActivity, -perc), y =
perc, fill = PhysicalActivity)) + geom_bar(stat = "identity", show.legend =
FALSE) + scale_fill_manual(values = smu2) + labs(x = "Physical Activity", y =
"Proportion with Heart Disease") + theme_minimal(base_family = "Open Sans")

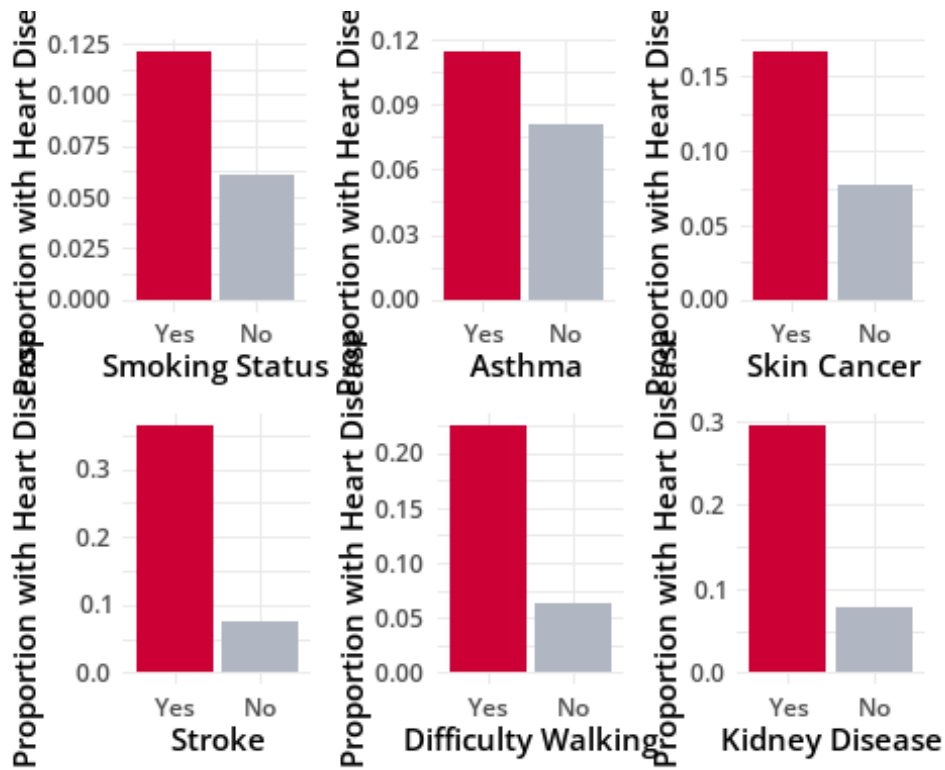
b7b <- ggplot(b7[c(3:4),], aes(x = reorder(Asthma, -perc), y = perc, fill =
Asthma)) + geom_bar(stat = "identity", show.legend = FALSE) +
scale_fill_manual(values = smu2) + labs(x = "Asthma", y = "Proportion with
Heart Disease") + theme_minimal(base_family = "Open Sans")

b8b <- ggplot(b8[c(3:4),], aes(x = reorder(KidneyDisease, -perc), y = perc,
fill = KidneyDisease)) + geom_bar(stat = "identity", show.legend = FALSE) +
scale_fill_manual(values = smu2) + labs(x = "Kidney Disease", y = "Proportion
with Heart Disease") + theme_minimal(base_family = "Open Sans")

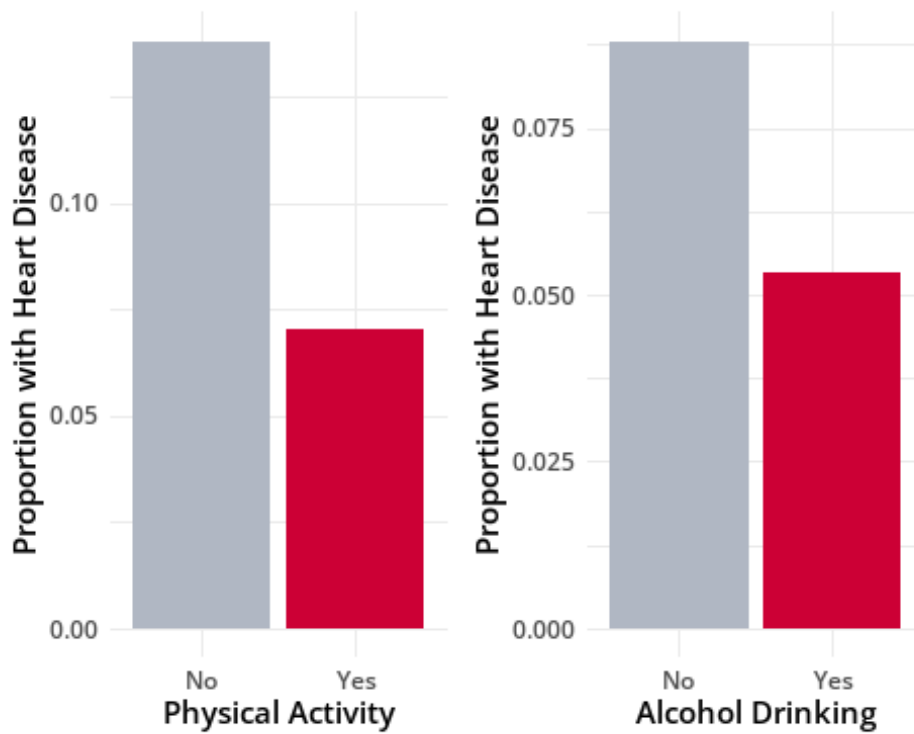
b9b <- ggplot(b9[c(3:4),], aes(x = reorder(SkinCancer, -perc), y = perc, fill
= SkinCancer)) +
  geom_bar(stat = "identity", show.legend = FALSE) + scale_fill_manual(values
= smu2) +
  labs(x = "Skin Cancer", y = "Proportion with Heart Disease") +
  theme_minimal(base_family = "Open Sans")

(b1b | b7b | b9b) / (b3b | b4b | b8b)

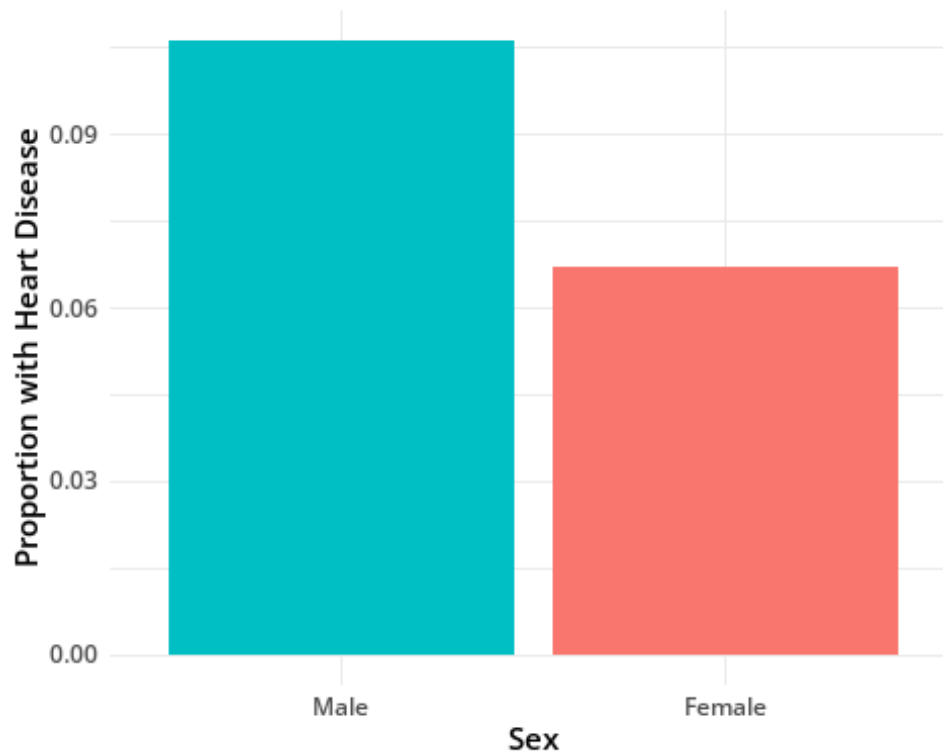
```



(b6b | b2b)



b5b

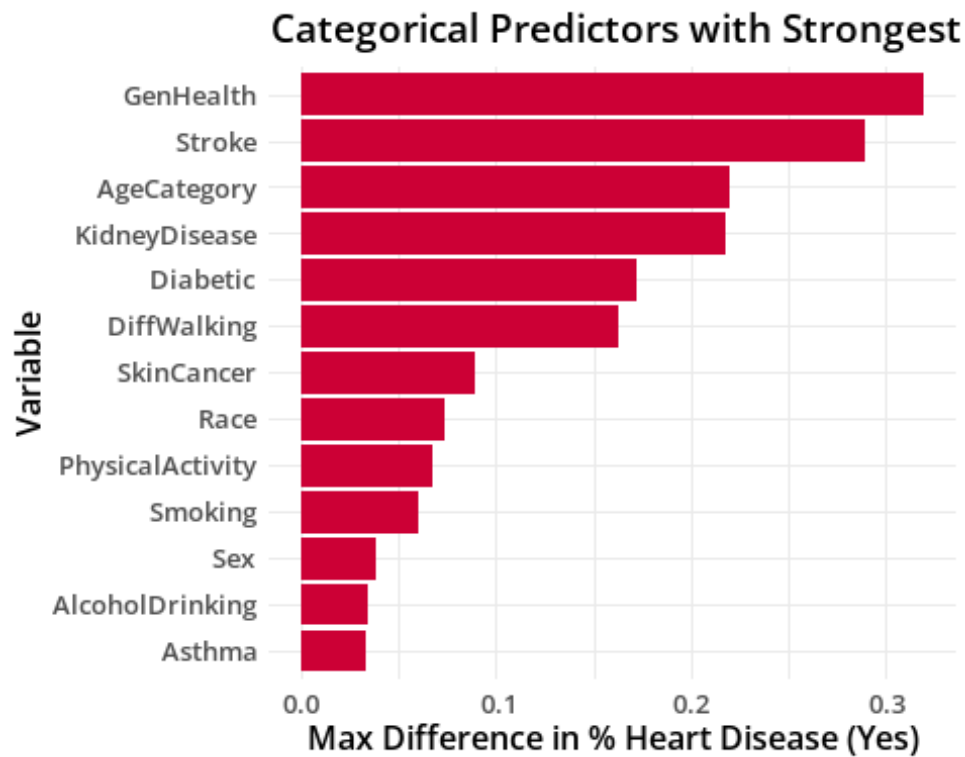


Categorical Variables: Heart Disease Proportions

```
cat_vars <- c("Smoking", "AlcoholDrinking", "Stroke",
"DiffWalking", "Sex", "AgeCategory", "Race", "Diabetic", "PhysicalActivity",
"GenHealth", "Asthma", "KidneyDisease", "SkinCancer")

library(purrr)
cat_var_stats <- map_dfr(cat_vars, function(v) {
  prop <- trainHeart %>% group_by(.data[[v]]) %>%
    summarize(p_heart = mean(HeartDisease == "Yes", na.rm = TRUE)) %>%
    summarize(max_diff = max(p_heart) - min(p_heart))
  tibble(var = v, max_diff = prop$max_diff)
})

cat_var_stats %>% arrange(desc(max_diff)) %>%
  slice_max(max_diff, n = 15) %>%
  mutate(var = fct_reorder(var, max_diff)) %>%
  ggplot(aes(max_diff, var)) + geom_col(fill = "#CC0035") + labs(x = "Max
Difference in % Heart Disease (Yes)", y = "Variable", title = "Categorical
Predictors with Strongest Variation") + theme_minimal(base_family = "Open
Sans", base_size = 12)
```

Scatterplot Matrix

`head(trainHeart)`

```
##   HeartDisease Smoking AlcoholDrinking Stroke DiffWalking Sex
## 1          No     Yes              No    No          No Female
## 2          No     No              No    Yes          No Female
## 3          No     Yes              No    No          No  Male
## 4          No     No              No    No          No Female
## 5          No     No              No    No          Yes Female
## 6          No     No              No    No          No Female
##   PhysicalActivity Asthma KidneyDisease SkinCancer Diabetic GenHealth
Race
## 1             Yes   Yes              No          Yes   Yes Very good
White
## 2             Yes   No              No          No     No Very good
White
## 3             Yes   Yes              No          No   Yes   Fair
White
## 4             No   No              No          Yes   No   Good
White
## 5             Yes   No              No          No   No Very good
White
## 6             Yes   Yes              No          Yes   No   Fair
White
##   AgeCategory BMI PhysicalHealth MentalHealth SleepTime GenHealth.num
## 1      55-59 16.60              3           30         5         4
```

```
## 2 80 or older 20.34      0      0      7      4
## 3      65-69 26.58      20     30      8      2
## 4      75-79 24.21      0      0      6      3
## 5      40-44 23.71      28      0      8      4
## 6      70-74 21.63      15      0      4      2
##   AgeCategory.num HeartDisease.num
## 1           8      0
## 2          13      0
## 3          10      0
## 4          12      0
## 5           5      0
## 6          11      0
```

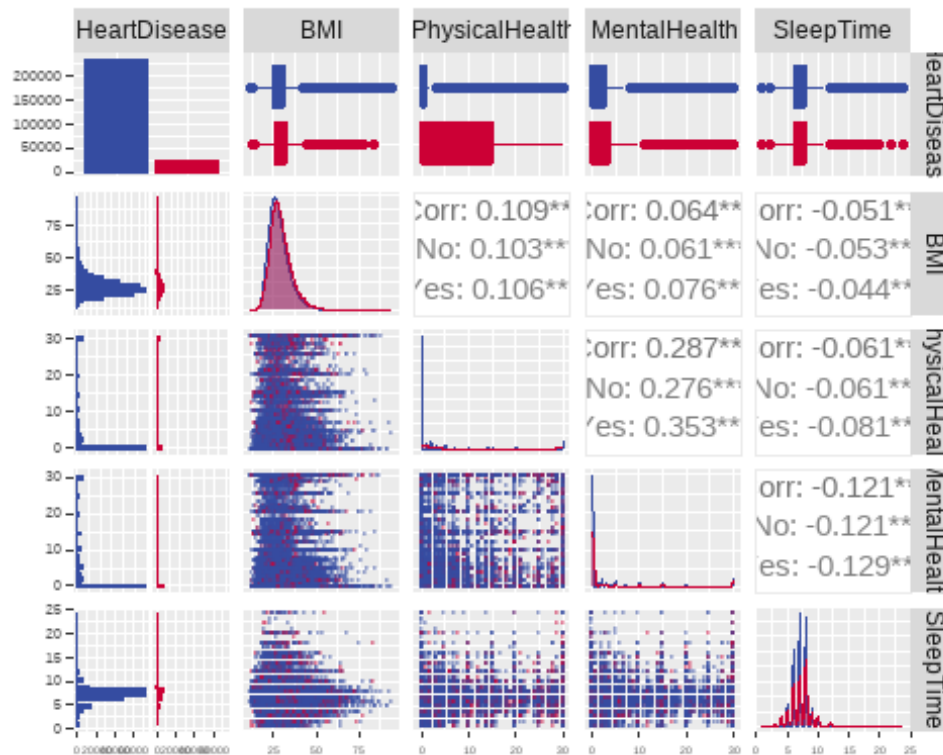
```
num.trainHeart = trainHeart[c(1,15:18)]
head(num.trainHeart)
```

```
##   HeartDisease   BMI PhysicalHealth MentalHealth SleepTime
## 1          No 16.60           3          30         5
## 2          No 20.34           0           0         7
## 3          No 26.58          20          30         8
## 4          No 24.21           0           0         6
## 5          No 23.71          28           0         8
## 6          No 21.63          15           0         4
```

```
num.trainHeart$HeartDisease <- factor(num.trainHeart$HeartDisease, levels =
c("No", "Yes"))
```

```
smu_colors <- c("No" = "#354CA1", "Yes" = "#CC0035")
p <- ggpairs(num.trainHeart, mapping = aes(color = HeartDisease, fill =
HeartDisease), lower = list(continuous = wrap("points", alpha = 0.3, size =
0.1)), diag = list(continuous = wrap("densityDiag", alpha = 0.4))) +
theme(axis.text.x = element_text(size = 4), axis.text.y = element_text(size =
5))
```

```
p + scale_color_manual(values = smu_colors) + scale_fill_manual(values =
smu_colors)
```



OBJECTIVE 1: Objective 1: Display the ability to perform EDA and build a logistic regression model for interpretation purposes using the training data. Perform your multiple logistic regression analysis and provide interpretation of the regression coefficients and confidence intervals for relevant predictor variables.

Logistic Regression Model

```
trainHeart$HeartDisease <- factor(trainHeart$HeartDisease, levels =
c("No", "Yes"))

logistic <- glm(HeartDisease~PhysicalHealth + BMI + AgeCategory + GenHealth +
Smoking + AlcoholDrinking + Stroke + DiffWalking + Sex + Race +Diabetic +
PhysicalActivity + Asthma + KidneyDisease + SkinCancer + MentalHealth +
SleepTime,data=trainHeart,family="binomial")
summary(logistic)

##
## Call:
## glm(formula = HeartDisease ~ PhysicalHealth + BMI + AgeCategory +
##     GenHealth + Smoking + AlcoholDrinking + Stroke + DiffWalking +
##     Sex + Race + Diabetic + PhysicalActivity + Asthma + KidneyDisease +
##     SkinCancer + MentalHealth + SleepTime, family = "binomial",
##     data = trainHeart)
##
```

```

## Coefficients:
##               Estimate Std. Error z value Pr(>|z|)
## (Intercept)    -6.2881623   0.1286529  -48.877 < 2e-16 ***
## PhysicalHealth  0.0032759   0.0009625   3.404 0.000665 ***
## BMI            0.0079718   0.0012810   6.223 4.88e-10 ***
## AgeCategory25-29 0.1553017   0.1383778   1.122 0.261734
## AgeCategory30-34 0.4808528   0.1248188   3.852 0.000117 ***
## AgeCategory35-39 0.5775600   0.1197558   4.823 1.42e-06 ***
## AgeCategory40-44 0.9949860   0.1123549   8.856 < 2e-16 ***
## AgeCategory45-49 1.3355719   0.1083289  12.329 < 2e-16 ***
## AgeCategory50-54 1.7571550   0.1045690  16.804 < 2e-16 ***
## AgeCategory55-59 1.9742682   0.1029841  19.171 < 2e-16 ***
## AgeCategory60-64 2.2439788   0.1020458  21.990 < 2e-16 ***
## AgeCategory65-69 2.4895451   0.1017218  24.474 < 2e-16 ***
## AgeCategory70-74 2.7752330   0.1016643  27.298 < 2e-16 ***
## AgeCategory75-79 2.9673379   0.1022894  29.009 < 2e-16 ***
## AgeCategory80 or older 3.2306427   0.1019929  31.675 < 2e-16 ***
## GenHealthFair    1.5373408   0.0366906  41.900 < 2e-16 ***
## GenHealthGood     1.0445423   0.0330597  31.596 < 2e-16 ***
## GenHealthPoor     1.9139797   0.0456757  41.904 < 2e-16 ***
## GenHealthVery good 0.4652281   0.0339338  13.710 < 2e-16 ***
## SmokingYes        0.3395607   0.0160667  21.134 < 2e-16 ***
## AlcoholDrinkingYes -0.2080333   0.0372214  -5.589 2.28e-08 ***
## StrokeYes         1.0463656   0.0252801  41.391 < 2e-16 ***
## DiffWalkingYes    0.2005539   0.0202855   9.887 < 2e-16 ***
## SexMale           0.7092266   0.0162828  43.557 < 2e-16 ***
## RaceAsian         -0.5806232   0.0944774  -6.146 7.96e-10 ***
## RaceBlack         -0.3777098   0.0643032  -5.874 4.26e-09 ***
## RaceHispanic      -0.2825286   0.0655127  -4.313 1.61e-05 ***
## RaceOther         -0.0748928   0.0712224  -1.052 0.293013
## RaceWhite         -0.1071057   0.0574076  -1.866 0.062082 .
## DiabeticNo, borderline diabetes 0.1209395   0.0471183   2.567 0.010267 *
## DiabeticYes        0.4652009   0.0187289  24.839 < 2e-16 ***
## DiabeticYes (during pregnancy) 0.2002676   0.1125688   1.779 0.075229 .
## PhysicalActivityYes 0.0145538   0.0179435   0.811 0.417313
## AsthmaYes         0.2695476   0.0215306  12.519 < 2e-16 ***
## KidneyDiseaseYes  0.5761332   0.0272599  21.135 < 2e-16 ***
## SkinCancerYes     0.1103143   0.0218634   5.046 4.52e-07 ***
## MentalHealth       0.0052765   0.0009863   5.350 8.81e-08 ***
## SleepTime         -0.0201204   0.0048546  -4.145 3.40e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 149527  on 255836  degrees of freedom
## Residual deviance: 116112  on 255799  degrees of freedom
## AIC: 116188
##
## Number of Fisher Scoring iterations: 7

```

```
AIC(logistic) # AIC: 116151
```

```
## [1] 116188.4
```

```
simple <- glm(HeartDisease~PhysicalHealth + BMI + AgeCategory + GenHealth +  
Smoking + AlcoholDrinking + Stroke + DiffWalking + Sex + Race +Diabetic +  
Asthma + KidneyDisease + SkinCancer + MentalHealth +  
SleepTime,data=trainHeart,family="binomial")
```

```
#summary(simple)
```

```
AIC(simple) # AIC: 115989.1
```

```
## [1] 116187.1
```

```
# Estimate for odds ratio
```

```
exp(cbind("Odds ratio" = coef(simple), confint.default(simple, level =  
0.95)))
```

##	Odds ratio	2.5 %	97.5 %
## (Intercept)	0.00188847	0.001472072	0.002422652
## PhysicalHealth	1.00323286	1.001345858	1.005123414
## BMI	1.00790935	1.005392175	1.010432829
## AgeCategory25-29	1.16827400	0.890755716	1.532254149
## AgeCategory30-34	1.61795215	1.266838260	2.066379934
## AgeCategory35-39	1.78200060	1.409199598	2.253425368
## AgeCategory40-44	2.70467587	2.170098939	3.370939192
## AgeCategory45-49	3.80145425	3.074262195	4.700657752
## AgeCategory50-54	5.79317070	4.719639271	7.110888116
## AgeCategory55-59	7.19809414	5.882458737	8.807976660
## AgeCategory60-64	9.42596025	7.717313705	11.512908516
## AgeCategory65-69	12.04992894	9.871898115	14.708497368
## AgeCategory70-74	16.03131241	13.135171301	19.566016442
## AgeCategory75-79	19.41869619	15.891296238	23.729075096
## AgeCategory80 or older	25.24731772	20.673921671	30.832420783
## GenHealthFair	4.64144943	4.320334856	4.986431262
## GenHealthGood	2.83836705	2.660490521	3.028136145
## GenHealthPoor	6.75742979	6.181042272	7.387565930
## GenHealthVery good	1.59196261	1.489530953	1.701438262
## SmokingYes	1.40345380	1.359996443	1.448299789
## AlcoholDrinkingYes	0.81206649	0.754932870	0.873524004
## StrokeYes	2.84702532	2.709401041	2.991640230
## DiffWalkingYes	1.21908572	1.172087349	1.267968634
## SexMale	2.03382108	1.970022719	2.099685516
## RaceAsian	0.55968253	0.465079434	0.673529114
## RaceBlack	0.68546398	0.604297144	0.777532835
## RaceHispanic	0.75334337	0.662574517	0.856547024
## RaceOther	0.92826407	0.807327326	1.067317016
## RaceWhite	0.89864701	0.803017567	1.005664741
## DiabeticNo, borderline diabetes	1.12885121	1.029271591	1.238064925
## DiabeticYes	1.59171552	1.534365995	1.651208582
## DiabeticYes (during pregnancy)	1.22222547	0.980238954	1.523949938
## AsthmaYes	1.30958472	1.255472995	1.366028690

```
## KidneyDiseaseYes          1.77867102  1.686140258  1.876279607
## SkinCancerYes            1.11734308  1.070504403  1.166231121
## MentalHealth             1.00526856  1.003327906  1.007212960
## SleepTime                0.98000511  0.970727693  0.989371199
```

#1 Unit increase odds ratio:

```
exp(0.0090482)
```

```
## [1] 1.009089
```

VIF

```
library(car)
```

```
vif(simple)
```

```
##              GVIF Df GVIF^(1/(2*Df))
## PhysicalHealth 1.798021 1      1.340903
## BMI           1.184161 1      1.088191
## AgeCategory   1.331403 12      1.011998
## GenHealth     1.944538 4       1.086681
## Smoking       1.053077 1       1.026195
## AlcoholDrinking 1.021417 1      1.010652
## Stroke        1.035732 1       1.017709
## DiffWalking   1.424655 1       1.193589
## Sex           1.076304 1       1.037451
## Race          1.098786 5       1.009465
## Diabetic      1.163092 3       1.025500
## Asthma        1.061801 1       1.030437
## KidneyDisease 1.050610 1       1.024992
## SkinCancer    1.082121 1       1.040250
## MentalHealth  1.244566 1       1.115601
## SleepTime     1.046353 1       1.022914
```

effect plots

```
library(sjPlot)
```

```
library(sjmisc)
```

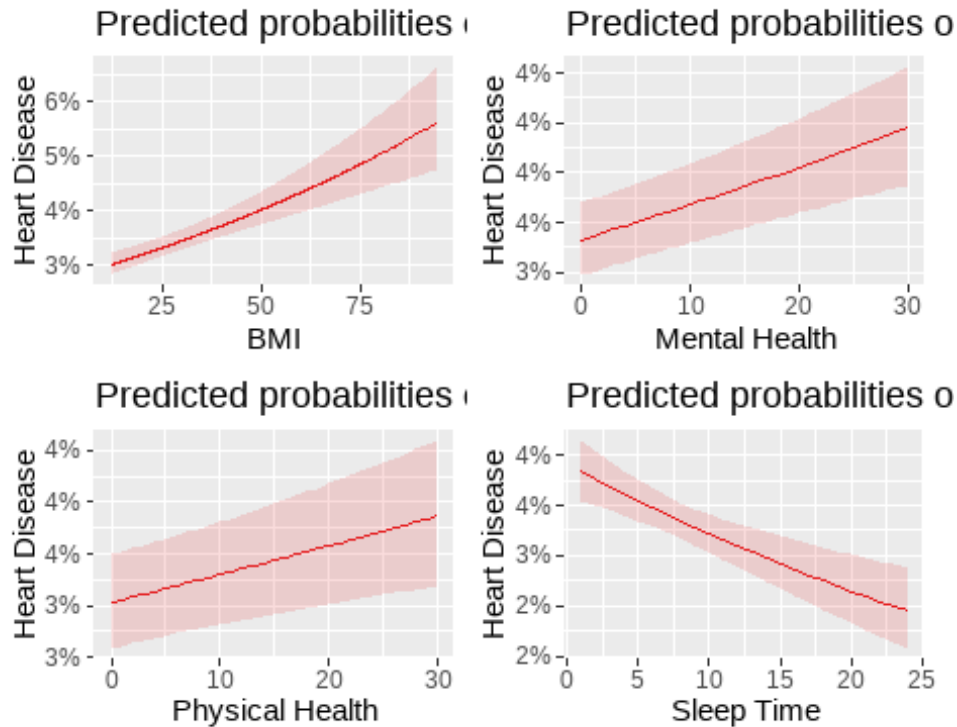
```
a1=plot_model(simple,type="pred",terms=c("BMI[all]"))
```

```
a2=plot_model(simple,type="pred",terms=c("MentalHealth[all]"))
```

```
a3=plot_model(simple,type="pred",terms=c("PhysicalHealth[all]"))
```

```
a4=plot_model(simple,type="pred",terms=c("SleepTime[all]"))
```

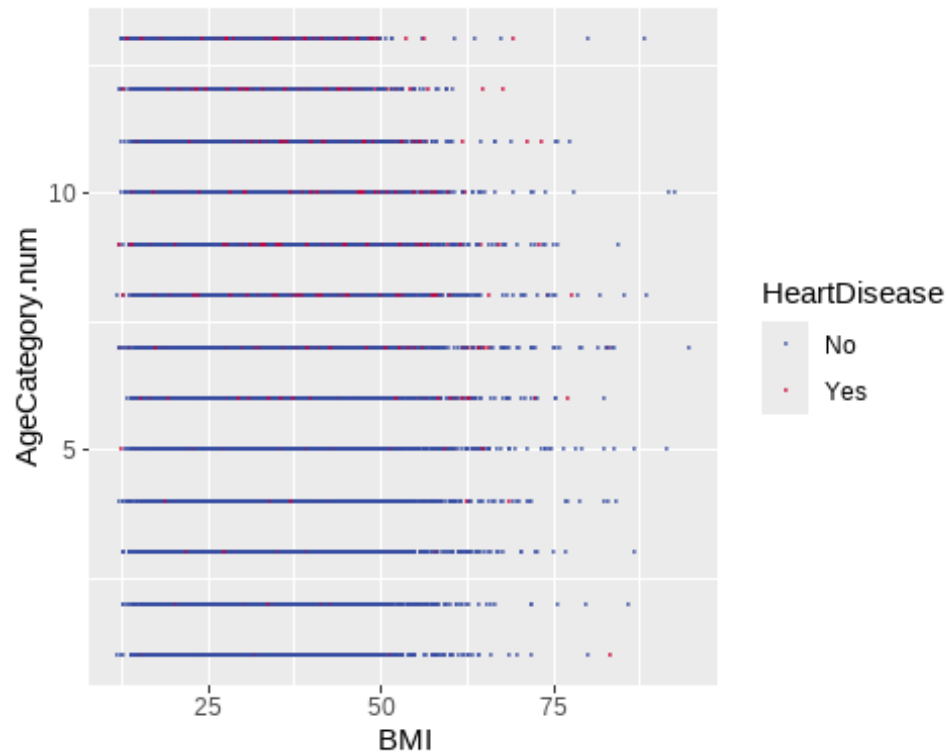
```
(a1|a2)/(a3|a4)
```



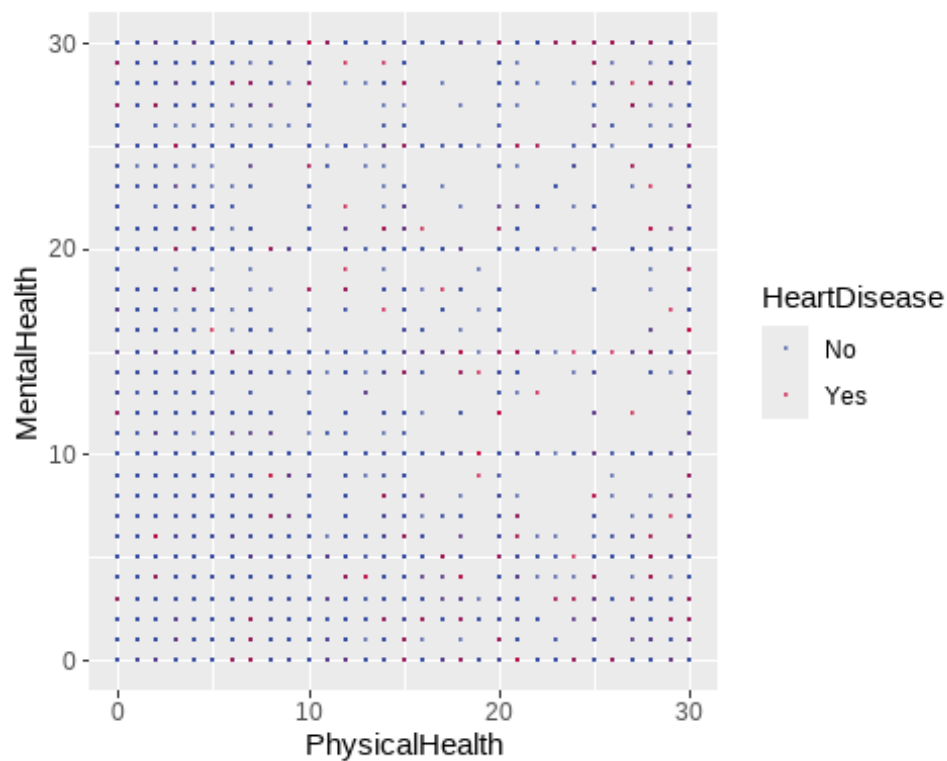
Objective 2: Build 4-5 additional classification models to compare to your model in objective 1. The goal of this objective is to build a model where prediction performance is prioritized.

Scatter Plots for Interactions

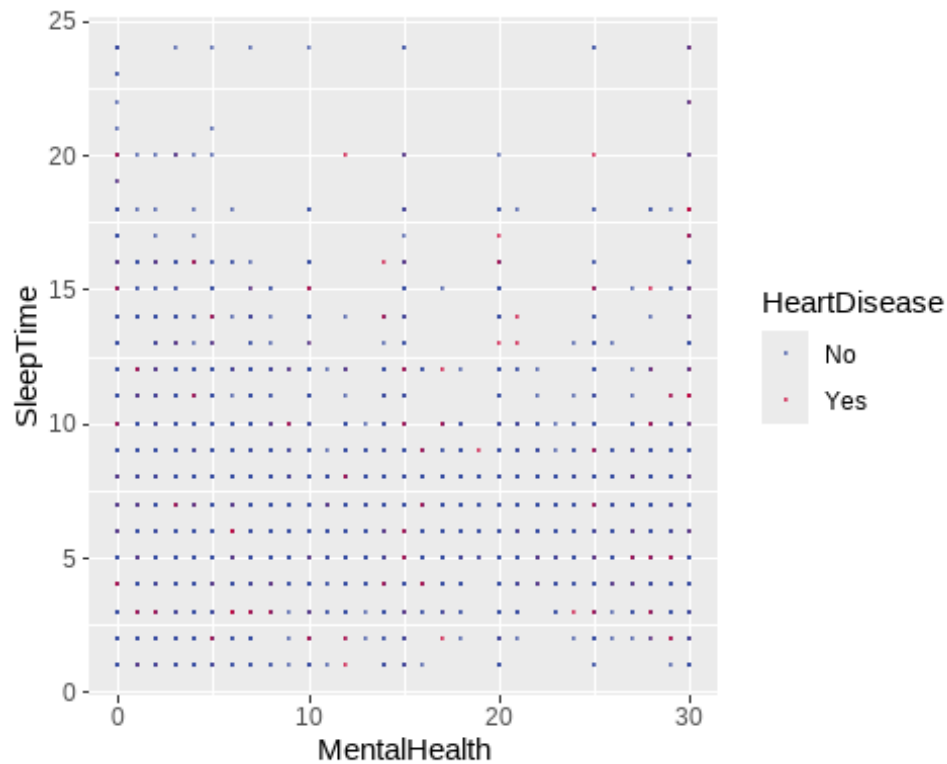
```
ggplot(trainHeart, aes(x=BMI, y=AgeCategory.num, color =
HeartDisease)) + geom_point(alpha = 0.5, size = 0.3) + scale_color_manual(values
= smu_colors)
```



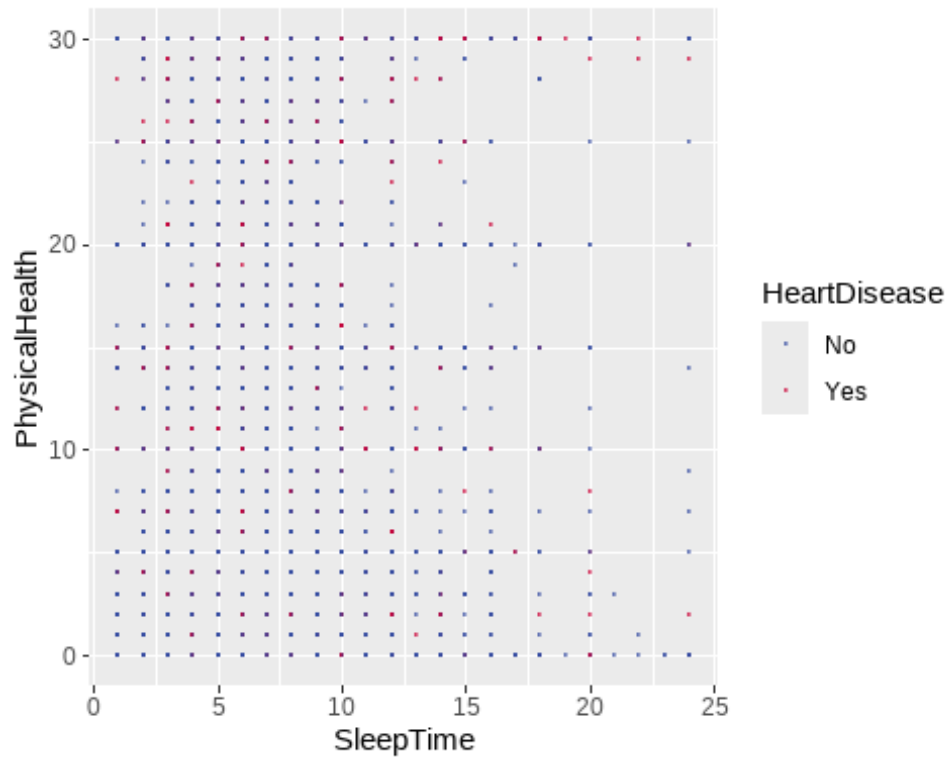
```
ggplot(trainHeart,aes(x=PhysicalHealth,y=MentalHealth, color =
HeartDisease))+geom_point(alpha = 0.5, size = 0.3)+ scale_color_manual(values
= smu_colors)
```




```
ggplot(trainHeart,aes(x=MentalHealth,y=SleepTime, color =
HeartDisease))+geom_point(alpha = 0.5, size = 0.3)+ scale_color_manual(values
= smu_colors)
```



```
ggplot(trainHeart,aes(x=SleepTime,y=PhysicalHealth, color =
HeartDisease))+geom_point(alpha = 0.5, size = 0.3)+ scale_color_manual(values
= smu_colors)
```



Feature Selection for Non-Additive Model

```
fitControl<-
trainControl(method="repeatedcv",number=5,repeats=1,classProbs=TRUE,
summaryFunction=mnLogLoss)
set.seed(1234)

# Feature Selection
logistic.fit1<-train(HeartDisease ~ GenHealth + AgeCategory + Sex + poly(BMI,
3) + poly(PhysicalHealth, 3) + poly(MentalHealth, 3) + poly(SleepTime, 3) +
poly(BMI, 3):GenHealth + poly(PhysicalHealth, 3):GenHealth +
poly(MentalHealth, 3):GenHealth + poly(SleepTime, 3):GenHealth + poly(BMI,
3):AgeCategory + poly(PhysicalHealth, 3):AgeCategory + poly(MentalHealth,
3):AgeCategory + poly(SleepTime, 3):AgeCategory + GenHealth:AgeCategory +
GenHealth:Sex + AgeCategory:Sex + Stroke + KidneyDisease + Diabetic +
DiffWalking + Smoking + AlcoholDrinking + Race + PhysicalActivity + Asthma +
SkinCancer,data=trainHeart,
method="glmnet",
trControl=fitControl,
metric="logLoss")
coef(logistic.fit1$finalModel,logistic.fit1$finalModel$lambdaOpt)

## 302 x 1 sparse Matrix of class "dgCMatrix"
##
## (Intercept) -5.082381e+00 s1
## GenHealthFair 1.365475e+00
## GenHealthGood 5.209784e-01
```

## GenHealthPoor	2.016405e+00
## GenHealthVery good	.
## AgeCategory25-29	-2.408000e-01
## AgeCategory30-34	.
## AgeCategory35-39	-1.481856e-02
## AgeCategory40-44	.
## AgeCategory45-49	1.062255e-01
## AgeCategory50-54	6.733045e-01
## AgeCategory55-59	7.449667e-01
## AgeCategory60-64	1.140479e+00
## AgeCategory65-69	1.252103e+00
## AgeCategory70-74	1.476053e+00
## AgeCategory75-79	1.840210e+00
## AgeCategory80 or older	2.419676e+00
## SexMale	1.341756e-01
## poly(BMI, 3)1	6.976063e+01
## poly(BMI, 3)2	.
## poly(BMI, 3)3	4.348724e-01
## poly(PhysicalHealth, 3)1	6.123639e+01
## poly(PhysicalHealth, 3)2	-2.230915e+01
## poly(PhysicalHealth, 3)3	1.778967e+00
## poly(MentalHealth, 3)1	1.190629e+01
## poly(MentalHealth, 3)2	1.103054e+01
## poly(MentalHealth, 3)3	.
## poly(SleepTime, 3)1	-2.974859e+00
## poly(SleepTime, 3)2	3.134767e+01
## poly(SleepTime, 3)3	-4.135554e+00
## StrokeYes	1.027050e+00
## KidneyDiseaseYes	5.660790e-01
## DiabeticNo, borderline diabetes	9.419706e-02
## DiabeticYes	4.506710e-01
## DiabeticYes (during pregnancy)	1.118347e-01
## DiffWalkingYes	1.965493e-01
## SmokingYes	3.267143e-01
## AlcoholDrinkingYes	-1.907485e-01
## RaceAsian	-4.294309e-01
## RaceBlack	-2.849335e-01
## RaceHispanic	-1.874744e-01
## RaceOther	.
## RaceWhite	-5.971448e-05
## PhysicalActivityYes	7.798310e-03
## AsthmaYes	2.333381e-01
## SkinCancerYes	1.044034e-01
## GenHealthFair:poly(BMI, 3)1	-7.539566e+01
## GenHealthGood:poly(BMI, 3)1	-6.303465e+01
## GenHealthPoor:poly(BMI, 3)1	-5.519019e+01
## GenHealthVery good:poly(BMI, 3)1	-4.611984e+01
## GenHealthFair:poly(BMI, 3)2	-2.813178e+01
## GenHealthGood:poly(BMI, 3)2	2.635939e+00
## GenHealthPoor:poly(BMI, 3)2	-2.317306e+00

## GenHealthVery good:poly(BMI, 3)2	5.105812e+00
## GenHealthFair:poly(BMI, 3)3	6.342999e+00
## GenHealthGood:poly(BMI, 3)3	.
## GenHealthPoor:poly(BMI, 3)3	-5.422949e+00
## GenHealthVery good:poly(BMI, 3)3	-7.305276e+00
## GenHealthFair:poly(PhysicalHealth, 3)1	-3.978254e+01
## GenHealthGood:poly(PhysicalHealth, 3)1	-5.152734e+01
## GenHealthPoor:poly(PhysicalHealth, 3)1	-5.426217e+01
## GenHealthVery good:poly(PhysicalHealth, 3)1	.
## GenHealthFair:poly(PhysicalHealth, 3)2	.
## GenHealthGood:poly(PhysicalHealth, 3)2	-2.935700e+00
## GenHealthPoor:poly(PhysicalHealth, 3)2	1.330828e+01
## GenHealthVery good:poly(PhysicalHealth, 3)2	.
## GenHealthFair:poly(PhysicalHealth, 3)3	-4.667020e+00
## GenHealthGood:poly(PhysicalHealth, 3)3	1.064579e+01
## GenHealthPoor:poly(PhysicalHealth, 3)3	-1.082578e+01
## GenHealthVery good:poly(PhysicalHealth, 3)3	.
## GenHealthFair:poly(MentalHealth, 3)1	-2.304104e+01
## GenHealthGood:poly(MentalHealth, 3)1	.
## GenHealthPoor:poly(MentalHealth, 3)1	-2.045942e+01
## GenHealthVery good:poly(MentalHealth, 3)1	-4.549592e-02
## GenHealthFair:poly(MentalHealth, 3)2	4.423024e-01
## GenHealthGood:poly(MentalHealth, 3)2	6.376921e+00
## GenHealthPoor:poly(MentalHealth, 3)2	.
## GenHealthVery good:poly(MentalHealth, 3)2	.
## GenHealthFair:poly(MentalHealth, 3)3	.
## GenHealthGood:poly(MentalHealth, 3)3	.
## GenHealthPoor:poly(MentalHealth, 3)3	2.192813e+00
## GenHealthVery good:poly(MentalHealth, 3)3	-1.607612e+01
## GenHealthFair:poly(SleepTime, 3)1	-7.201402e+00
## GenHealthGood:poly(SleepTime, 3)1	.
## GenHealthPoor:poly(SleepTime, 3)1	-1.918736e+00
## GenHealthVery good:poly(SleepTime, 3)1	-1.139107e+01
## GenHealthFair:poly(SleepTime, 3)2	-1.155977e+01
## GenHealthGood:poly(SleepTime, 3)2	-6.149964e+00
## GenHealthPoor:poly(SleepTime, 3)2	-1.144382e+01
## GenHealthVery good:poly(SleepTime, 3)2	.
## GenHealthFair:poly(SleepTime, 3)3	2.462735e+00
## GenHealthGood:poly(SleepTime, 3)3	.
## GenHealthPoor:poly(SleepTime, 3)3	7.426681e+00
## GenHealthVery good:poly(SleepTime, 3)3	-2.180389e+00
## AgeCategory25-29:poly(BMI, 3)1	-3.881652e+01
## AgeCategory30-34:poly(BMI, 3)1	-1.149409e+01
## AgeCategory35-39:poly(BMI, 3)1	.
## AgeCategory40-44:poly(BMI, 3)1	3.393162e+01
## AgeCategory45-49:poly(BMI, 3)1	2.801001e+01
## AgeCategory50-54:poly(BMI, 3)1	.
## AgeCategory55-59:poly(BMI, 3)1	1.953293e+01
## AgeCategory60-64:poly(BMI, 3)1	2.654297e+01
## AgeCategory65-69:poly(BMI, 3)1	3.000840e+01

## AgeCategory70-74:poly(BMI, 3)1	1.813365e+01
## AgeCategory75-79:poly(BMI, 3)1	1.562326e+01
## AgeCategory80 or older:poly(BMI, 3)1	-4.051895e+00
## AgeCategory25-29:poly(BMI, 3)2	.
## AgeCategory30-34:poly(BMI, 3)2	.
## AgeCategory35-39:poly(BMI, 3)2	.
## AgeCategory40-44:poly(BMI, 3)2	.
## AgeCategory45-49:poly(BMI, 3)2	.
## AgeCategory50-54:poly(BMI, 3)2	1.438277e+01
## AgeCategory55-59:poly(BMI, 3)2	-2.343817e+01
## AgeCategory60-64:poly(BMI, 3)2	-2.292692e+01
## AgeCategory65-69:poly(BMI, 3)2	-2.966274e+01
## AgeCategory70-74:poly(BMI, 3)2	-8.062572e+00
## AgeCategory75-79:poly(BMI, 3)2	1.354845e+01
## AgeCategory80 or older:poly(BMI, 3)2	-1.551391e+01
## AgeCategory25-29:poly(BMI, 3)3	-6.845376e+01
## AgeCategory30-34:poly(BMI, 3)3	-6.312503e+00
## AgeCategory35-39:poly(BMI, 3)3	.
## AgeCategory40-44:poly(BMI, 3)3	-8.468877e+00
## AgeCategory45-49:poly(BMI, 3)3	5.759447e+00
## AgeCategory50-54:poly(BMI, 3)3	-9.403213e+00
## AgeCategory55-59:poly(BMI, 3)3	.
## AgeCategory60-64:poly(BMI, 3)3	-2.563235e+00
## AgeCategory65-69:poly(BMI, 3)3	7.015881e+00
## AgeCategory70-74:poly(BMI, 3)3	1.186066e+01
## AgeCategory75-79:poly(BMI, 3)3	4.336842e+01
## AgeCategory80 or older:poly(BMI, 3)3	1.050027e+01
## AgeCategory25-29:poly(PhysicalHealth, 3)1	2.617794e+01
## AgeCategory30-34:poly(PhysicalHealth, 3)1	6.421398e+00
## AgeCategory35-39:poly(PhysicalHealth, 3)1	1.793761e+01
## AgeCategory40-44:poly(PhysicalHealth, 3)1	6.307178e+01
## AgeCategory45-49:poly(PhysicalHealth, 3)1	3.645794e+01
## AgeCategory50-54:poly(PhysicalHealth, 3)1	2.358233e+00
## AgeCategory55-59:poly(PhysicalHealth, 3)1	1.053838e+01
## AgeCategory60-64:poly(PhysicalHealth, 3)1	-1.961290e-01
## AgeCategory65-69:poly(PhysicalHealth, 3)1	3.185611e+00
## AgeCategory70-74:poly(PhysicalHealth, 3)1	-3.224334e+01
## AgeCategory75-79:poly(PhysicalHealth, 3)1	-5.606065e+00
## AgeCategory80 or older:poly(PhysicalHealth, 3)1	.
## AgeCategory25-29:poly(PhysicalHealth, 3)2	.
## AgeCategory30-34:poly(PhysicalHealth, 3)2	-4.865232e+00
## AgeCategory35-39:poly(PhysicalHealth, 3)2	-5.496160e+00
## AgeCategory40-44:poly(PhysicalHealth, 3)2	-1.864174e+01
## AgeCategory45-49:poly(PhysicalHealth, 3)2	5.162134e+00
## AgeCategory50-54:poly(PhysicalHealth, 3)2	.
## AgeCategory55-59:poly(PhysicalHealth, 3)2	-1.247563e+01
## AgeCategory60-64:poly(PhysicalHealth, 3)2	1.268972e-01
## AgeCategory65-69:poly(PhysicalHealth, 3)2	-2.978012e+00
## AgeCategory70-74:poly(PhysicalHealth, 3)2	2.612426e+00
## AgeCategory75-79:poly(PhysicalHealth, 3)2	-1.145318e+00

```

## AgeCategory80 or older:poly(PhysicalHealth, 3)2 4.514250e-02
## AgeCategory25-29:poly(PhysicalHealth, 3)3 -6.802217e+00
## AgeCategory30-34:poly(PhysicalHealth, 3)3 5.010016e+00
## AgeCategory35-39:poly(PhysicalHealth, 3)3 -1.853067e+01
## AgeCategory40-44:poly(PhysicalHealth, 3)3 9.450329e-01
## AgeCategory45-49:poly(PhysicalHealth, 3)3 -8.192603e+00
## AgeCategory50-54:poly(PhysicalHealth, 3)3 .
## AgeCategory55-59:poly(PhysicalHealth, 3)3 8.295898e+00
## AgeCategory60-64:poly(PhysicalHealth, 3)3 1.735066e+01
## AgeCategory65-69:poly(PhysicalHealth, 3)3 4.975433e-01
## AgeCategory70-74:poly(PhysicalHealth, 3)3 1.150897e+01
## AgeCategory75-79:poly(PhysicalHealth, 3)3 9.207291e+00
## AgeCategory80 or older:poly(PhysicalHealth, 3)3 6.536431e+00
## AgeCategory25-29:poly(MentalHealth, 3)1 .
## AgeCategory30-34:poly(MentalHealth, 3)1 1.221309e+01
## AgeCategory35-39:poly(MentalHealth, 3)1 .
## AgeCategory40-44:poly(MentalHealth, 3)1 4.752234e+01
## AgeCategory45-49:poly(MentalHealth, 3)1 2.928571e+01
## AgeCategory50-54:poly(MentalHealth, 3)1 2.839820e+01
## AgeCategory55-59:poly(MentalHealth, 3)1 1.720801e+01
## AgeCategory60-64:poly(MentalHealth, 3)1 .
## AgeCategory65-69:poly(MentalHealth, 3)1 1.207154e+00
## AgeCategory70-74:poly(MentalHealth, 3)1 5.497186e+00
## AgeCategory75-79:poly(MentalHealth, 3)1 1.875255e+01
## AgeCategory80 or older:poly(MentalHealth, 3)1 -1.557383e+01
## AgeCategory25-29:poly(MentalHealth, 3)2 1.442164e+01
## AgeCategory30-34:poly(MentalHealth, 3)2 -4.624028e+01
## AgeCategory35-39:poly(MentalHealth, 3)2 -3.989787e+01
## AgeCategory40-44:poly(MentalHealth, 3)2 4.421333e-01
## AgeCategory45-49:poly(MentalHealth, 3)2 1.532661e-01
## AgeCategory50-54:poly(MentalHealth, 3)2 -1.958209e+01
## AgeCategory55-59:poly(MentalHealth, 3)2 -4.985132e-01
## AgeCategory60-64:poly(MentalHealth, 3)2 -2.368318e+01
## AgeCategory65-69:poly(MentalHealth, 3)2 -4.461539e+00
## AgeCategory70-74:poly(MentalHealth, 3)2 -2.199899e+00
## AgeCategory75-79:poly(MentalHealth, 3)2 5.410209e-01
## AgeCategory80 or older:poly(MentalHealth, 3)2 1.207026e+00
## AgeCategory25-29:poly(MentalHealth, 3)3 .
## AgeCategory30-34:poly(MentalHealth, 3)3 4.644122e+00
## AgeCategory35-39:poly(MentalHealth, 3)3 .
## AgeCategory40-44:poly(MentalHealth, 3)3 -9.477994e+00
## AgeCategory45-49:poly(MentalHealth, 3)3 -1.311213e+01
## AgeCategory50-54:poly(MentalHealth, 3)3 -2.374229e+00
## AgeCategory55-59:poly(MentalHealth, 3)3 -2.075420e+00
## AgeCategory60-64:poly(MentalHealth, 3)3 .
## AgeCategory65-69:poly(MentalHealth, 3)3 -2.658872e+00
## AgeCategory70-74:poly(MentalHealth, 3)3 -2.667174e+00
## AgeCategory75-79:poly(MentalHealth, 3)3 2.530578e+00
## AgeCategory80 or older:poly(MentalHealth, 3)3 -2.063662e+01
## AgeCategory25-29:poly(SleepTime, 3)1 -3.288146e+01

```

```

## AgeCategory30-34:poly(SleepTime, 3)1 .
## AgeCategory35-39:poly(SleepTime, 3)1 -1.971758e+01
## AgeCategory40-44:poly(SleepTime, 3)1 -5.665283e+01
## AgeCategory45-49:poly(SleepTime, 3)1 -1.229973e+01
## AgeCategory50-54:poly(SleepTime, 3)1 -5.385573e+00
## AgeCategory55-59:poly(SleepTime, 3)1 -5.702650e+00
## AgeCategory60-64:poly(SleepTime, 3)1 -6.992544e-01
## AgeCategory65-69:poly(SleepTime, 3)1 -6.305977e+00
## AgeCategory70-74:poly(SleepTime, 3)1 .
## AgeCategory75-79:poly(SleepTime, 3)1 -7.116107e+00
## AgeCategory80 or older:poly(SleepTime, 3)1 1.017115e+00
## AgeCategory25-29:poly(SleepTime, 3)2 1.467663e+00
## AgeCategory30-34:poly(SleepTime, 3)2 .
## AgeCategory35-39:poly(SleepTime, 3)2 2.945451e+01
## AgeCategory40-44:poly(SleepTime, 3)2 .
## AgeCategory45-49:poly(SleepTime, 3)2 -3.691595e+00
## AgeCategory50-54:poly(SleepTime, 3)2 2.994464e+00
## AgeCategory55-59:poly(SleepTime, 3)2 .
## AgeCategory60-64:poly(SleepTime, 3)2 -8.814033e+00
## AgeCategory65-69:poly(SleepTime, 3)2 -4.255532e+00
## AgeCategory70-74:poly(SleepTime, 3)2 -2.189092e+01
## AgeCategory75-79:poly(SleepTime, 3)2 -1.244383e+01
## AgeCategory80 or older:poly(SleepTime, 3)2 -2.217425e+01
## AgeCategory25-29:poly(SleepTime, 3)3 -5.999650e+01
## AgeCategory30-34:poly(SleepTime, 3)3 -3.783766e+01
## AgeCategory35-39:poly(SleepTime, 3)3 -3.406648e+01
## AgeCategory40-44:poly(SleepTime, 3)3 -1.948987e+01
## AgeCategory45-49:poly(SleepTime, 3)3 -3.918162e+01
## AgeCategory50-54:poly(SleepTime, 3)3 -4.969500e+00
## AgeCategory55-59:poly(SleepTime, 3)3 -6.563903e+00
## AgeCategory60-64:poly(SleepTime, 3)3 -1.178077e+01
## AgeCategory65-69:poly(SleepTime, 3)3 .
## AgeCategory70-74:poly(SleepTime, 3)3 5.040294e+00
## AgeCategory75-79:poly(SleepTime, 3)3 -6.590724e+00
## AgeCategory80 or older:poly(SleepTime, 3)3 1.119539e+01
## GenHealthFair:AgeCategory25-29 1.216232e-02
## GenHealthGood:AgeCategory25-29 .
## GenHealthPoor:AgeCategory25-29 -3.253621e-01
## GenHealthVery good:AgeCategory25-29 -5.264035e-02
## GenHealthFair:AgeCategory30-34 .
## GenHealthGood:AgeCategory30-34 -1.046955e-01
## GenHealthPoor:AgeCategory30-34 1.000804e-01
## GenHealthVery good:AgeCategory30-34 -1.562067e-02
## GenHealthFair:AgeCategory35-39 2.726501e-01
## GenHealthGood:AgeCategory35-39 .
## GenHealthPoor:AgeCategory35-39 .
## GenHealthVery good:AgeCategory35-39 .
## GenHealthFair:AgeCategory40-44 2.602604e-01
## GenHealthGood:AgeCategory40-44 2.700057e-01
## GenHealthPoor:AgeCategory40-44 -2.491170e-02

```

## GenHealthVery good:AgeCategory40-44	.
## GenHealthFair:AgeCategory45-49	3.850686e-01
## GenHealthGood:AgeCategory45-49	5.018100e-01
## GenHealthPoor:AgeCategory45-49	4.025937e-01
## GenHealthVery good:AgeCategory45-49	.
## GenHealthFair:AgeCategory50-54	3.528804e-01
## GenHealthGood:AgeCategory50-54	4.688302e-01
## GenHealthPoor:AgeCategory50-54	2.105372e-01
## GenHealthVery good:AgeCategory50-54	3.947218e-02
## GenHealthFair:AgeCategory55-59	4.044764e-01
## GenHealthGood:AgeCategory55-59	5.899475e-01
## GenHealthPoor:AgeCategory55-59	2.593376e-01
## GenHealthVery good:AgeCategory55-59	2.588464e-01
## GenHealthFair:AgeCategory60-64	1.398233e-01
## GenHealthGood:AgeCategory60-64	5.067101e-01
## GenHealthPoor:AgeCategory60-64	2.519584e-02
## GenHealthVery good:AgeCategory60-64	1.287890e-01
## GenHealthFair:AgeCategory65-69	2.241767e-01
## GenHealthGood:AgeCategory65-69	4.740614e-01
## GenHealthPoor:AgeCategory65-69	1.029397e-02
## GenHealthVery good:AgeCategory65-69	3.941552e-01
## GenHealthFair:AgeCategory70-74	3.452994e-01
## GenHealthGood:AgeCategory70-74	5.578432e-01
## GenHealthPoor:AgeCategory70-74	2.033519e-01
## GenHealthVery good:AgeCategory70-74	4.580431e-01
## GenHealthFair:AgeCategory75-79	-6.178645e-03
## GenHealthGood:AgeCategory75-79	4.184162e-01
## GenHealthPoor:AgeCategory75-79	-1.570126e-01
## GenHealthVery good:AgeCategory75-79	3.629403e-01
## GenHealthFair:AgeCategory80 or older	-2.228558e-01
## GenHealthGood:AgeCategory80 or older	1.967705e-01
## GenHealthPoor:AgeCategory80 or older	-5.627838e-01
## GenHealthVery good:AgeCategory80 or older	1.172062e-01
## GenHealthFair:SexMale	-1.371995e-01
## GenHealthGood:SexMale	-2.466067e-03
## GenHealthPoor:SexMale	-1.845542e-01
## GenHealthVery good:SexMale	2.232157e-01
## AgeCategory25-29:SexMale	-4.950389e-02
## AgeCategory30-34:SexMale	2.509475e-02
## AgeCategory35-39:SexMale	.
## AgeCategory40-44:SexMale	2.359291e-01
## AgeCategory45-49:SexMale	5.208506e-01
## AgeCategory50-54:SexMale	4.672873e-01
## AgeCategory55-59:SexMale	4.982574e-01
## AgeCategory60-64:SexMale	5.911650e-01
## AgeCategory65-69:SexMale	6.859425e-01
## AgeCategory70-74:SexMale	6.903051e-01
## AgeCategory75-79:SexMale	6.856742e-01
## AgeCategory80 or older:SexMale	4.587236e-01


```

# Feature Selection After Dropping Features Above
logistic.fit2 <- train(HeartDisease ~ poly(BMI, 2) + poly(PhysicalHealth, 2)
+ poly(MentalHealth, 2) + poly(SleepTime, 3) + poly(BMI, 2):GenHealth +
poly(PhysicalHealth, 2):GenHealth + poly(MentalHealth, 2):GenHealth +
poly(SleepTime, 2):GenHealth + poly(BMI, 2):AgeCategory +
poly(PhysicalHealth, 2):AgeCategory + poly(MentalHealth, 2):AgeCategory +
poly(SleepTime, 3):AgeCategory + GenHealth:AgeCategory + GenHealth:Sex +
AgeCategory:Sex + AgeCategory + GenHealth + Smoking + AlcoholDrinking +
Stroke + DiffWalking + Sex + Race + Diabetic + PhysicalActivity + Asthma +
KidneyDisease + SkinCancer,
  data = trainHeart,
  method = "glmnet",
  trControl = fitControl,
  metric = "logLoss")

logistic.fit2

## glmnet
##
## 255837 samples
##      17 predictor
##      2 classes: 'No', 'Yes'
##
## No pre-processing
## Resampling: Cross-Validated (5 fold, repeated 1 times)
## Summary of sample sizes: 204670, 204670, 204669, 204669, 204670
## Resampling results across tuning parameters:
##
##   alpha  lambda      logLoss
##   0.10   0.000112293  0.2262822
##   0.10   0.001122930  0.2264407
##   0.10   0.011229304  0.2286249
##   0.55   0.000112293  0.2262453
##   0.55   0.001122930  0.2267719
##   0.55   0.011229304  0.2351449
##   1.00   0.000112293  0.2262426
##   1.00   0.001122930  0.2273043
##   1.00   0.011229304  0.2415035
##
## logLoss was used to select the optimal model using the smallest value.
## The final values used for the model were alpha = 1 and lambda =
0.000112293.

coef(logistic.fit2$finalModel, logistic.fit2$finalModel$lambdaOpt)

## 249 x 1 sparse Matrix of class "dgCMatrix"
##
## (Intercept) -5.078651e+00
## poly(BMI, 2)1 7.253900e+01
## poly(BMI, 2)2 .

```

## poly(PhysicalHealth, 2)1	6.118200e+01
## poly(PhysicalHealth, 2)2	-2.271905e+01
## poly(MentalHealth, 2)1	1.186894e+01
## poly(MentalHealth, 2)2	1.031408e+01
## poly(SleepTime, 3)1	-3.641896e+00
## poly(SleepTime, 3)2	1.948736e+01
## poly(SleepTime, 3)3	-6.327816e+00
## AgeCategory25-29	-1.939562e-01
## AgeCategory30-34	.
## AgeCategory35-39	-1.118358e-02
## AgeCategory40-44	.
## AgeCategory45-49	1.027848e-01
## AgeCategory50-54	6.697000e-01
## AgeCategory55-59	7.403101e-01
## AgeCategory60-64	1.133206e+00
## AgeCategory65-69	1.253731e+00
## AgeCategory70-74	1.473888e+00
## AgeCategory75-79	1.840972e+00
## AgeCategory80 or older	2.425435e+00
## GenHealthFair	1.345594e+00
## GenHealthGood	5.132796e-01
## GenHealthPoor	1.939595e+00
## GenHealthVery good	.
## SmokingYes	3.261232e-01
## AlcoholDrinkingYes	-1.915498e-01
## StrokeYes	1.026555e+00
## DiffWalkingYes	1.969704e-01
## SexMale	1.331510e-01
## RaceAsian	-4.313939e-01
## RaceBlack	-2.833738e-01
## RaceHispanic	-1.873330e-01
## RaceOther	.
## RaceWhite	-2.971840e-05
## DiabeticNo, borderline diabetes	9.271971e-02
## DiabeticYes	4.507525e-01
## DiabeticYes (during pregnancy)	1.121420e-01
## PhysicalActivityYes	8.016324e-03
## AsthmaYes	2.343960e-01
## KidneyDiseaseYes	5.669070e-01
## SkinCancerYes	1.051088e-01
## poly(BMI, 2)1:GenHealthFair	-7.784875e+01
## poly(BMI, 2)2:GenHealthFair	-3.054601e+01
## poly(BMI, 2)1:GenHealthGood	-6.414241e+01
## poly(BMI, 2)2:GenHealthGood	2.761301e-01
## poly(BMI, 2)1:GenHealthPoor	-5.495699e+01
## poly(BMI, 2)2:GenHealthPoor	-4.384504e+00
## poly(BMI, 2)1:GenHealthVery good	-4.702081e+01
## poly(BMI, 2)2:GenHealthVery good	2.635735e+00
## poly(PhysicalHealth, 2)1:GenHealthFair	-4.079962e+01
## poly(PhysicalHealth, 2)2:GenHealthFair	.

## poly(PhysicalHealth, 2)1:GenHealthGood	-5.352771e+01
## poly(PhysicalHealth, 2)2:GenHealthGood	-4.604887e+00
## poly(PhysicalHealth, 2)1:GenHealthPoor	-5.248148e+01
## poly(PhysicalHealth, 2)2:GenHealthPoor	9.842193e+00
## poly(PhysicalHealth, 2)1:GenHealthVery good	.
## poly(PhysicalHealth, 2)2:GenHealthVery good	.
## poly(MentalHealth, 2)1:GenHealthFair	-2.259751e+01
## poly(MentalHealth, 2)2:GenHealthFair	.
## poly(MentalHealth, 2)1:GenHealthGood	.
## poly(MentalHealth, 2)2:GenHealthGood	5.511848e+00
## poly(MentalHealth, 2)1:GenHealthPoor	-1.990624e+01
## poly(MentalHealth, 2)2:GenHealthPoor	-6.177655e-01
## poly(MentalHealth, 2)1:GenHealthVery good	-3.780370e-01
## poly(MentalHealth, 2)2:GenHealthVery good	.
## GenHealthExcellent:poly(SleepTime, 2)1	.
## GenHealthFair:poly(SleepTime, 2)1	-6.681484e+00
## GenHealthGood:poly(SleepTime, 2)1	.
## GenHealthPoor:poly(SleepTime, 2)1	.
## GenHealthVery good:poly(SleepTime, 2)1	-1.042676e+01
## GenHealthExcellent:poly(SleepTime, 2)2	2.467371e+01
## GenHealthFair:poly(SleepTime, 2)2	-5.452166e-01
## GenHealthGood:poly(SleepTime, 2)2	2.312382e+00
## GenHealthPoor:poly(SleepTime, 2)2	.
## GenHealthVery good:poly(SleepTime, 2)2	7.619722e+00
## poly(BMI, 2)1:AgeCategory25-29	-3.658933e+01
## poly(BMI, 2)2:AgeCategory25-29	.
## poly(BMI, 2)1:AgeCategory30-34	-1.137029e+01
## poly(BMI, 2)2:AgeCategory30-34	.
## poly(BMI, 2)1:AgeCategory35-39	.
## poly(BMI, 2)2:AgeCategory35-39	.
## poly(BMI, 2)1:AgeCategory40-44	3.361882e+01
## poly(BMI, 2)2:AgeCategory40-44	-6.456087e-01
## poly(BMI, 2)1:AgeCategory45-49	2.464469e+01
## poly(BMI, 2)2:AgeCategory45-49	4.648304e+00
## poly(BMI, 2)1:AgeCategory50-54	.
## poly(BMI, 2)2:AgeCategory50-54	1.042270e+01
## poly(BMI, 2)1:AgeCategory55-59	1.746561e+01
## poly(BMI, 2)2:AgeCategory55-59	-2.167157e+01
## poly(BMI, 2)1:AgeCategory60-64	2.512776e+01
## poly(BMI, 2)2:AgeCategory60-64	-2.090836e+01
## poly(BMI, 2)1:AgeCategory65-69	2.644109e+01
## poly(BMI, 2)2:AgeCategory65-69	-2.975377e+01
## poly(BMI, 2)1:AgeCategory70-74	1.460175e+01
## poly(BMI, 2)2:AgeCategory70-74	-1.002306e+01
## poly(BMI, 2)1:AgeCategory75-79	4.611634e+00
## poly(BMI, 2)2:AgeCategory75-79	-7.699165e+00
## poly(BMI, 2)1:AgeCategory80 or older	-4.731753e+00
## poly(BMI, 2)2:AgeCategory80 or older	-1.710097e+01
## poly(PhysicalHealth, 2)1:AgeCategory25-29	2.908424e+01
## poly(PhysicalHealth, 2)2:AgeCategory25-29	.

```

## poly(PhysicalHealth, 2)1:AgeCategory30-34 7.945924e+00
## poly(PhysicalHealth, 2)2:AgeCategory30-34 -5.267223e+00
## poly(PhysicalHealth, 2)1:AgeCategory35-39 2.668604e+01
## poly(PhysicalHealth, 2)2:AgeCategory35-39 -1.106581e+01
## poly(PhysicalHealth, 2)1:AgeCategory40-44 6.678245e+01
## poly(PhysicalHealth, 2)2:AgeCategory40-44 -1.991180e+01
## poly(PhysicalHealth, 2)1:AgeCategory45-49 3.901888e+01
## poly(PhysicalHealth, 2)2:AgeCategory45-49 9.230968e-01
## poly(PhysicalHealth, 2)1:AgeCategory50-54 3.076684e+00
## poly(PhysicalHealth, 2)2:AgeCategory50-54 -1.796556e-01
## poly(PhysicalHealth, 2)1:AgeCategory55-59 9.933182e+00
## poly(PhysicalHealth, 2)2:AgeCategory55-59 -1.023054e+01
## poly(PhysicalHealth, 2)1:AgeCategory60-64 -9.974266e-01
## poly(PhysicalHealth, 2)2:AgeCategory60-64 4.432250e+00
## poly(PhysicalHealth, 2)1:AgeCategory65-69 4.031612e+00
## poly(PhysicalHealth, 2)2:AgeCategory65-69 -3.180032e+00
## poly(PhysicalHealth, 2)1:AgeCategory70-74 -3.195537e+01
## poly(PhysicalHealth, 2)2:AgeCategory70-74 3.837492e+00
## poly(PhysicalHealth, 2)1:AgeCategory75-79 -5.456134e+00
## poly(PhysicalHealth, 2)2:AgeCategory75-79 .
## poly(PhysicalHealth, 2)1:AgeCategory80 or older .
## poly(PhysicalHealth, 2)2:AgeCategory80 or older 9.292088e-01
## poly(MentalHealth, 2)1:AgeCategory25-29 5.085271e-01
## poly(MentalHealth, 2)2:AgeCategory25-29 1.487111e+01
## poly(MentalHealth, 2)1:AgeCategory30-34 1.309619e+01
## poly(MentalHealth, 2)2:AgeCategory30-34 -4.492247e+01
## poly(MentalHealth, 2)1:AgeCategory35-39 7.029462e-02
## poly(MentalHealth, 2)2:AgeCategory35-39 -3.748721e+01
## poly(MentalHealth, 2)1:AgeCategory40-44 4.911658e+01
## poly(MentalHealth, 2)2:AgeCategory40-44 .
## poly(MentalHealth, 2)1:AgeCategory45-49 3.002037e+01
## poly(MentalHealth, 2)2:AgeCategory45-49 .
## poly(MentalHealth, 2)1:AgeCategory50-54 2.874427e+01
## poly(MentalHealth, 2)2:AgeCategory50-54 -1.931318e+01
## poly(MentalHealth, 2)1:AgeCategory55-59 1.699767e+01
## poly(MentalHealth, 2)2:AgeCategory55-59 .
## poly(MentalHealth, 2)1:AgeCategory60-64 .
## poly(MentalHealth, 2)2:AgeCategory60-64 -2.223269e+01
## poly(MentalHealth, 2)1:AgeCategory65-69 4.939290e-01
## poly(MentalHealth, 2)2:AgeCategory65-69 -3.125638e+00
## poly(MentalHealth, 2)1:AgeCategory70-74 4.621370e+00
## poly(MentalHealth, 2)2:AgeCategory70-74 -6.200590e-01
## poly(MentalHealth, 2)1:AgeCategory75-79 1.814275e+01
## poly(MentalHealth, 2)2:AgeCategory75-79 1.922470e+00
## poly(MentalHealth, 2)1:AgeCategory80 or older -1.863637e+01
## poly(MentalHealth, 2)2:AgeCategory80 or older 3.304723e+00
## poly(SleepTime, 3)1:AgeCategory25-29 -3.280830e+01
## poly(SleepTime, 3)2:AgeCategory25-29 1.357602e+00
## poly(SleepTime, 3)3:AgeCategory25-29 -5.604882e+01
## poly(SleepTime, 3)1:AgeCategory30-34 .

```

## poly(SleepTime, 3)2:AgeCategory30-34	.
## poly(SleepTime, 3)3:AgeCategory30-34	-3.456705e+01
## poly(SleepTime, 3)1:AgeCategory35-39	-2.048277e+01
## poly(SleepTime, 3)2:AgeCategory35-39	2.909808e+01
## poly(SleepTime, 3)3:AgeCategory35-39	-3.224611e+01
## poly(SleepTime, 3)1:AgeCategory40-44	-5.722544e+01
## poly(SleepTime, 3)2:AgeCategory40-44	.
## poly(SleepTime, 3)3:AgeCategory40-44	-1.553359e+01
## poly(SleepTime, 3)1:AgeCategory45-49	-1.148977e+01
## poly(SleepTime, 3)2:AgeCategory45-49	-2.724382e+00
## poly(SleepTime, 3)3:AgeCategory45-49	-3.438801e+01
## poly(SleepTime, 3)1:AgeCategory50-54	-5.386214e+00
## poly(SleepTime, 3)2:AgeCategory50-54	2.672838e+00
## poly(SleepTime, 3)3:AgeCategory50-54	-1.513931e+00
## poly(SleepTime, 3)1:AgeCategory55-59	-5.809355e+00
## poly(SleepTime, 3)2:AgeCategory55-59	.
## poly(SleepTime, 3)3:AgeCategory55-59	-1.399505e+00
## poly(SleepTime, 3)1:AgeCategory60-64	-7.077813e-01
## poly(SleepTime, 3)2:AgeCategory60-64	-7.562410e+00
## poly(SleepTime, 3)3:AgeCategory60-64	-7.539824e+00
## poly(SleepTime, 3)1:AgeCategory65-69	-6.496256e+00
## poly(SleepTime, 3)2:AgeCategory65-69	-4.362169e+00
## poly(SleepTime, 3)3:AgeCategory65-69	.
## poly(SleepTime, 3)1:AgeCategory70-74	.
## poly(SleepTime, 3)2:AgeCategory70-74	-2.093449e+01
## poly(SleepTime, 3)3:AgeCategory70-74	8.463982e+00
## poly(SleepTime, 3)1:AgeCategory75-79	-7.413216e+00
## poly(SleepTime, 3)2:AgeCategory75-79	-1.208403e+01
## poly(SleepTime, 3)3:AgeCategory75-79	-4.214708e+00
## poly(SleepTime, 3)1:AgeCategory80 or older	1.127135e+00
## poly(SleepTime, 3)2:AgeCategory80 or older	-2.000129e+01
## poly(SleepTime, 3)3:AgeCategory80 or older	1.585296e+01
## GenHealthFair:AgeCategory25-29	2.902427e-02
## GenHealthGood:AgeCategory25-29	.
## GenHealthPoor:AgeCategory25-29	-2.149868e-01
## GenHealthVery good:AgeCategory25-29	-7.148227e-02
## GenHealthFair:AgeCategory30-34	.
## GenHealthGood:AgeCategory30-34	-9.444820e-02
## GenHealthPoor:AgeCategory30-34	1.512698e-01
## GenHealthVery good:AgeCategory30-34	-1.920013e-02
## GenHealthFair:AgeCategory35-39	2.670775e-01
## GenHealthGood:AgeCategory35-39	.
## GenHealthPoor:AgeCategory35-39	.
## GenHealthVery good:AgeCategory35-39	.
## GenHealthFair:AgeCategory40-44	2.651192e-01
## GenHealthGood:AgeCategory40-44	2.760765e-01
## GenHealthPoor:AgeCategory40-44	.
## GenHealthVery good:AgeCategory40-44	.
## GenHealthFair:AgeCategory45-49	3.943451e-01
## GenHealthGood:AgeCategory45-49	5.064343e-01

## GenHealthPoor:AgeCategory45-49	4.593275e-01
## GenHealthVery good:AgeCategory45-49	.
## GenHealthFair:AgeCategory50-54	3.765812e-01
## GenHealthGood:AgeCategory50-54	4.810162e-01
## GenHealthPoor:AgeCategory50-54	2.846471e-01
## GenHealthVery good:AgeCategory50-54	3.687773e-02
## GenHealthFair:AgeCategory55-59	4.278975e-01
## GenHealthGood:AgeCategory55-59	6.041014e-01
## GenHealthPoor:AgeCategory55-59	3.317048e-01
## GenHealthVery good:AgeCategory55-59	2.573335e-01
## GenHealthFair:AgeCategory60-64	1.678912e-01
## GenHealthGood:AgeCategory60-64	5.261734e-01
## GenHealthPoor:AgeCategory60-64	9.991113e-02
## GenHealthVery good:AgeCategory60-64	1.309321e-01
## GenHealthFair:AgeCategory65-69	2.338629e-01
## GenHealthGood:AgeCategory65-69	4.769339e-01
## GenHealthPoor:AgeCategory65-69	6.403673e-02
## GenHealthVery good:AgeCategory65-69	3.878448e-01
## GenHealthFair:AgeCategory70-74	3.625682e-01
## GenHealthGood:AgeCategory70-74	5.660133e-01
## GenHealthPoor:AgeCategory70-74	2.630293e-01
## GenHealthVery good:AgeCategory70-74	4.547172e-01
## GenHealthFair:AgeCategory75-79	.
## GenHealthGood:AgeCategory75-79	4.200880e-01
## GenHealthPoor:AgeCategory75-79	-1.054591e-01
## GenHealthVery good:AgeCategory75-79	3.585139e-01
## GenHealthFair:AgeCategory80 or older	-2.099543e-01
## GenHealthGood:AgeCategory80 or older	1.989944e-01
## GenHealthPoor:AgeCategory80 or older	-5.029596e-01
## GenHealthVery good:AgeCategory80 or older	1.154595e-01
## GenHealthFair:SexMale	-1.322066e-01
## GenHealthGood:SexMale	-1.770511e-03
## GenHealthPoor:SexMale	-1.814723e-01
## GenHealthVery good:SexMale	2.253798e-01
## AgeCategory25-29:SexMale	-8.572142e-02
## AgeCategory30-34:SexMale	2.335956e-02
## AgeCategory35-39:SexMale	.
## AgeCategory40-44:SexMale	2.322237e-01
## AgeCategory45-49:SexMale	5.243804e-01
## AgeCategory50-54:SexMale	4.588320e-01
## AgeCategory55-59:SexMale	4.959286e-01
## AgeCategory60-64:SexMale	5.871558e-01
## AgeCategory65-69:SexMale	6.869518e-01
## AgeCategory70-74:SexMale	6.923235e-01
## AgeCategory75-79:SexMale	6.912564e-01
## AgeCategory80 or older:SexMale	4.615603e-01

Complex Logistic Regression Model

Additive Model for Comparison

```
simple <- glm(HeartDisease~PhysicalHealth + BMI + AgeCategory + GenHealth +  
Smoking + AlcoholDrinking + Stroke + DiffWalking + Sex + Race +Diabetic +  
Asthma + KidneyDisease + SkinCancer + MentalHealth + SleepTime +  
PhysicalActivity ,data=trainHeart,family="binomial")
```

Complex 1:

```
complex1 <- glm(HeartDisease ~ GenHealth + AgeCategory + Sex +  
GenHealth:AgeCategory + GenHealth:BMI + AgeCategory:Sex + Stroke +  
KidneyDisease + Diabetic + DiffWalking + Smoking + AlcoholDrinking + Race +  
PhysicalActivity + Asthma + SkinCancer + BMI + PhysicalHealth + MentalHealth  
+ SleepTime,data = trainHeart, family = "binomial")
```

Complex 2:

```
complex2 <- glm(HeartDisease ~ poly(BMI, 2) + poly(PhysicalHealth, 2) +  
poly(MentalHealth, 2) + poly(SleepTime, 3) + poly(BMI, 2):GenHealth +  
poly(PhysicalHealth, 2):GenHealth + poly(MentalHealth, 2):GenHealth +  
poly(SleepTime, 2):GenHealth + poly(BMI, 2):AgeCategory +  
poly(PhysicalHealth, 2):AgeCategory + poly(MentalHealth, 2):AgeCategory +  
poly(SleepTime, 3):AgeCategory + GenHealth:AgeCategory + GenHealth:Sex +  
AgeCategory:Sex + AgeCategory + GenHealth + Smoking + AlcoholDrinking +  
Stroke + DiffWalking + Sex + Race + Diabetic + PhysicalActivity + Asthma +  
KidneyDisease + SkinCancer, data = trainHeart, family = "binomial")
```

```
AIC(simple)
```

```
## [1] 116188.4
```

```
AIC(complex1)
```

```
## [1] 115858.2
```

```
AIC(complex2)
```

```
## [1] 115676.5
```

effect plots

```
library(sjPlot)
```

```
library(sjmisc)
```

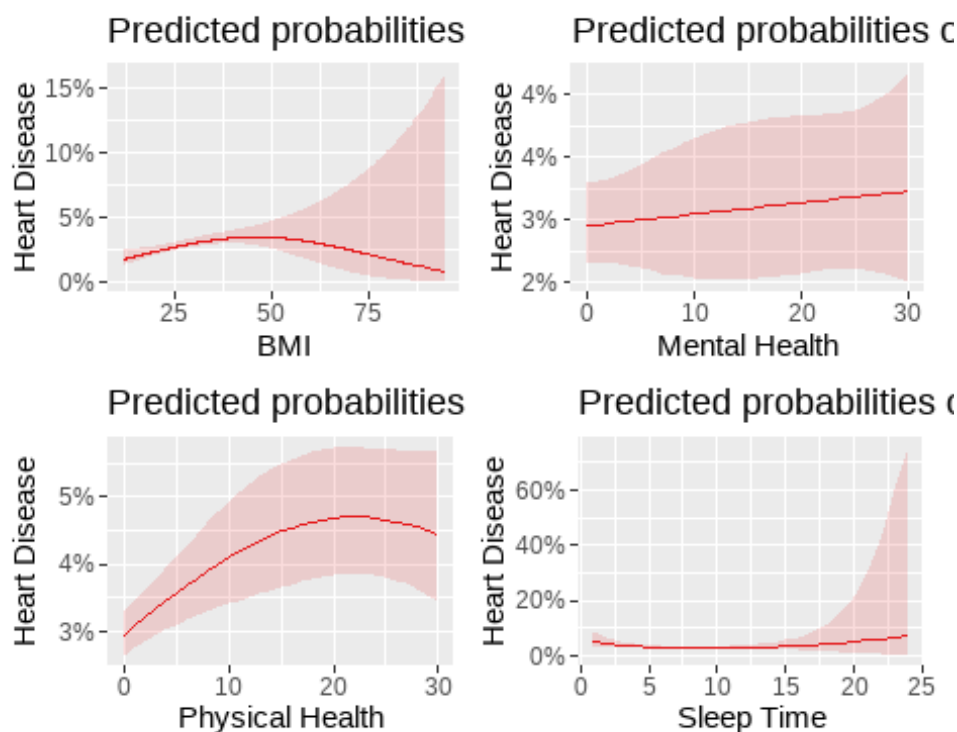
```
a5=plot_model(complex2,type="pred",terms=c("BMI[all]"))
```

```
a6=plot_model(complex2,type="pred",terms=c("MentalHealth[all]"))
```

```
a7=plot_model(complex2,type="pred",terms=c("PhysicalHealth[all]"))
```

```
a8=plot_model(complex2,type="pred",terms=c("SleepTime[all]"))
```

```
(a5|a6)/(a7|a8)
```



Sampling for KNN/QDA/LDA

```
library(pROC)
```

```
#Sample 10%
```

```
set.seed(1234)
```

```
sample_pct <- 0.10
```

```
train_sample_size <- floor(nrow(trainHeart) * sample_pct)
```

```
test_sample_size <- floor(nrow(testHeart) * sample_pct)
```

```
train_indices <- sample(seq_len(nrow(trainHeart)), size = train_sample_size)
```

```
test_indices <- sample(seq_len(nrow(testHeart)), size = test_sample_size)
```

```
trainHeart_sample <- trainHeart[train_indices, ]
```

```
testHeart_sample <- testHeart[test_indices, ]
```

```
#Identify numeric columns present in BOTH sets
```

```
numeric_cols <- intersect(
  names(trainHeart_sample)[sapply(trainHeart_sample, is.numeric)],
  names(testHeart_sample)[sapply(testHeart_sample, is.numeric)]
)
```

```
#Extract only shared numeric columns from both sets
```

```
train_numeric <- trainHeart_sample[, numeric_cols]
```

```
test_numeric <- testHeart_sample[, numeric_cols]
```



```

#Use preProcess to compute scaling from training data only
scale_params <- preProcess(train_numeric, method = c("center", "scale"))

#Apply scaling to both train and test
train_scaled <- predict(scale_params, train_numeric)
test_scaled <- predict(scale_params, test_numeric)

#Add back the target variable
train_scaled$HeartDisease <- factor(trainHeart_sample$HeartDisease, levels =
c("No", "Yes"))
test_scaled$HeartDisease <- factor(testHeart_sample$HeartDisease, levels =
c("No", "Yes"))

#Model Control
fitControl <- trainControl(method = "repeatedcv",
                           number = 5,
                           repeats = 1,
                           classProbs = TRUE,
                           summaryFunction = mnLogLoss)

```

KNN/LDA/QDA Models

```

#KNN Model
set.seed(1234)
grid <- expand.grid(k = c(3,5,7))
knn.fit <- train(HeartDisease ~ PhysicalHealth + BMI + MentalHealth +
SleepTime,
                data = train_scaled,
                method = "knn",
                trControl = fitControl,
                metric = "logLoss",
                tuneGrid = grid)

#LDA Model
lda.fit <- train(HeartDisease ~ PhysicalHealth + BMI + MentalHealth +
SleepTime,
                data = train_scaled,
                method = "lda",
                trControl = fitControl,
                metric = "logLoss")

#QDA Model
qda.fit <- train(HeartDisease ~ PhysicalHealth + BMI + MentalHealth +
SleepTime,
                data = train_scaled,
                method = "qda",
                trControl = fitControl,
                metric = "logLoss")

```

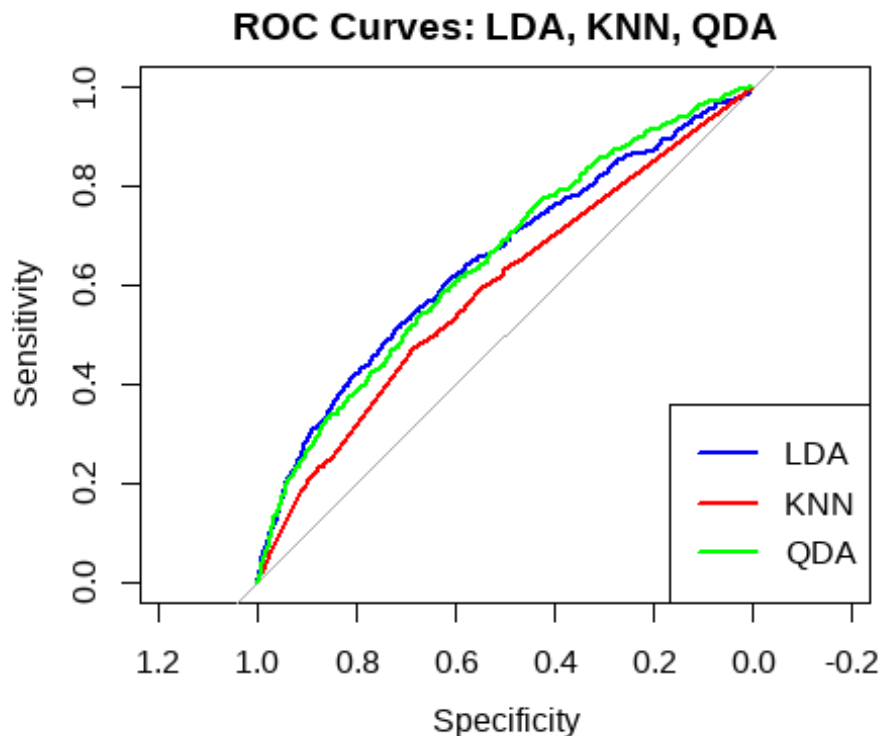
ROC/AUC

#ROC and AUC

```
actual <- test_scaled$HeartDisease
pred.knn <- predict(knn.fit, test_scaled, type = "prob")[, "Yes"]
pred.lda <- predict(lda.fit, test_scaled, type = "prob")[, "Yes"]
pred.qda <- predict(qda.fit, test_scaled, type = "prob")[, "Yes"]

roc.knn <- roc(actual, pred.knn, levels = c("No", "Yes"))
roc.lda <- roc(actual, pred.lda, levels = c("No", "Yes"))
roc.qda <- roc(actual, pred.qda, levels = c("No", "Yes"))

plot(roc.lda, col = "blue", lwd = 2, main = "ROC Curves: LDA, KNN, QDA")
plot(roc.knn, col = "red", lwd = 2, add = TRUE)
plot(roc.qda, col = "green", lwd = 2, add = TRUE)
legend("bottomright", legend = c("LDA", "KNN", "QDA"),
      col = c("blue", "red", "green"), lwd = 2)
```



```
cat("AUC LDA:", auc(roc.lda), "\n")
## AUC LDA: 0.6459434

cat("AUC KNN:", auc(roc.knn), "\n")
## AUC KNN: 0.5901466

cat("AUC QDA:", auc(roc.qda), "\n")
```

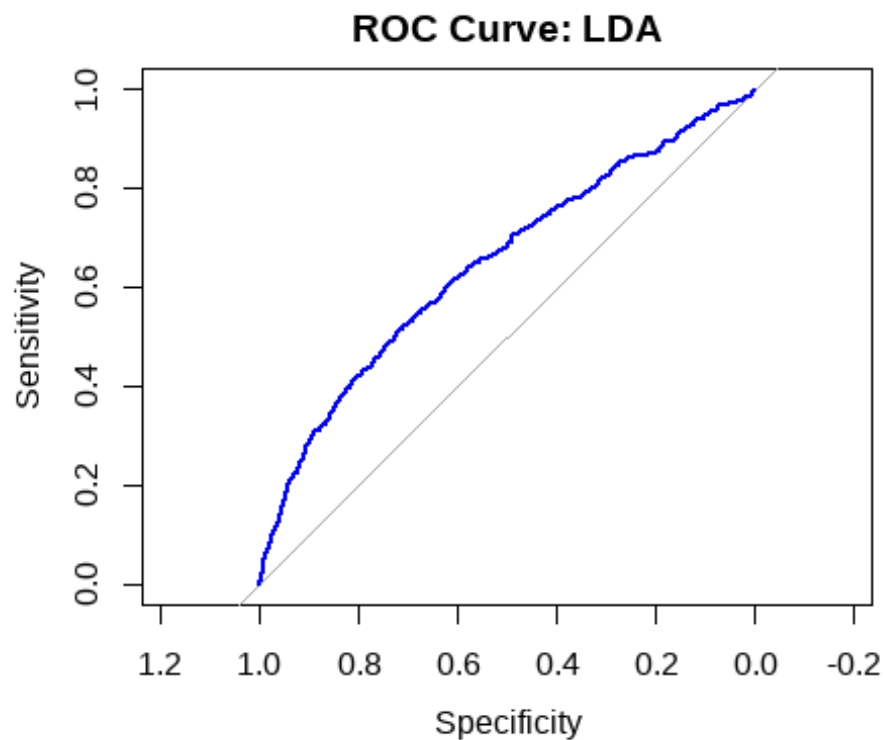
```
## AUC QDA: 0.6484842
```

```
actual <- test_scaled$HeartDisease
```

```
#Visualize More Plots
```

```
# LDA
```

```
pred.lda <- predict(lda.fit, test_scaled, type = "prob")[, "Yes"]  
roc.lda <- roc(actual, pred.lda, levels = c("No", "Yes"))  
plot(roc.lda, col = "blue", lwd = 2, main = "ROC Curve: LDA")
```

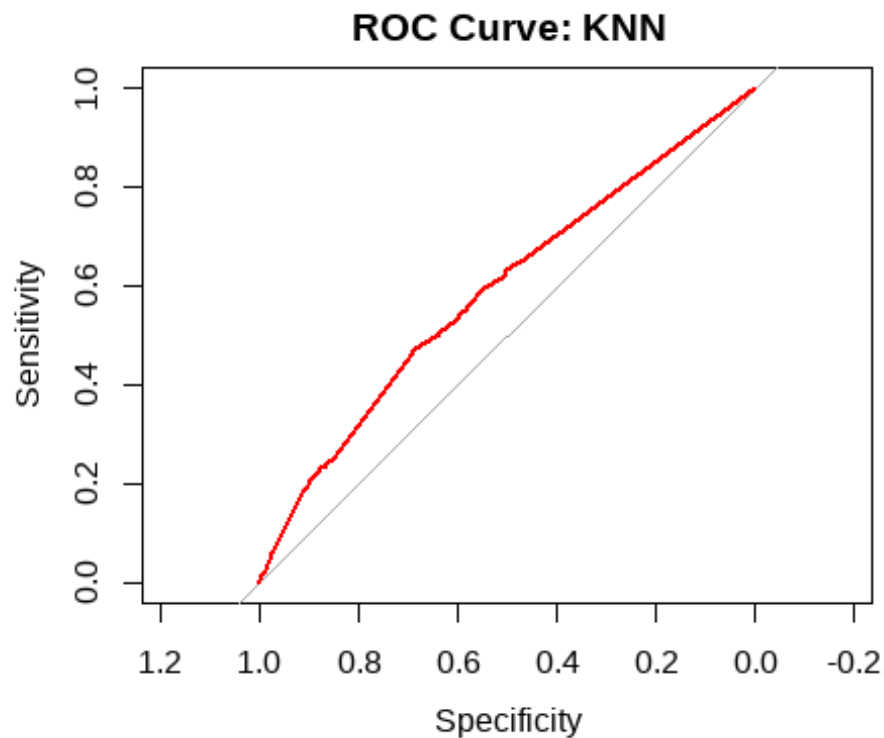


```
cat("AUC LDA:", auc(roc.lda), "\n")
```

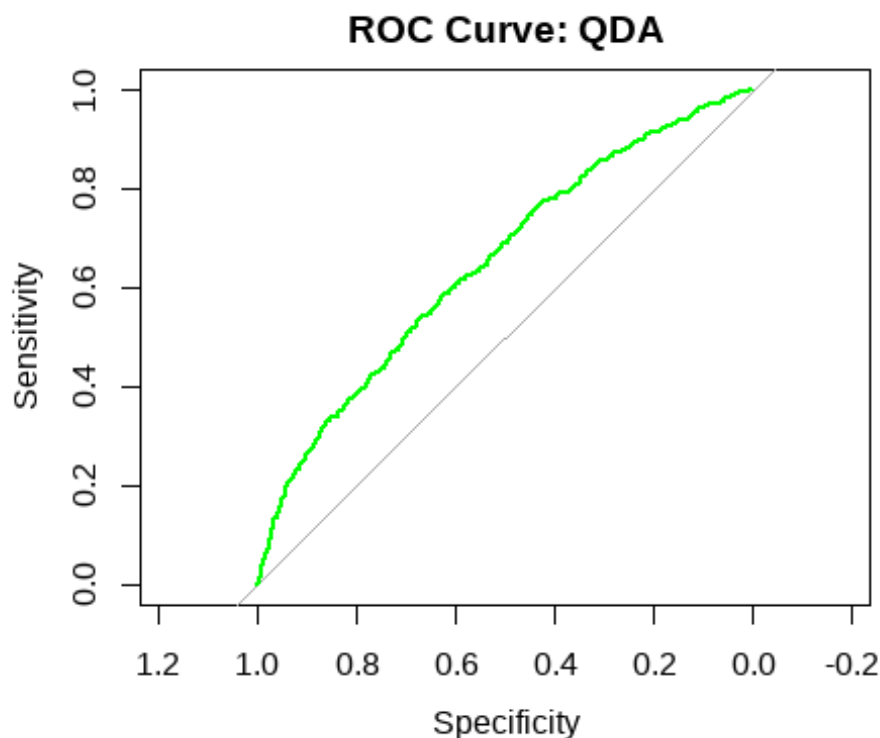
```
## AUC LDA: 0.6459434
```

```
# KNN
```

```
pred.knn <- predict(knn.fit, test_scaled, type = "prob")[, "Yes"]  
roc.knn <- roc(actual, pred.knn, levels = c("No", "Yes"))  
plot(roc.knn, col = "red", lwd = 2, main = "ROC Curve: KNN")
```



```
cat("AUC KNN:", auc(roc.knn), "\n")  
  
## AUC KNN: 0.5901466  
  
# QDA  
pred.qda <- predict(qda.fit, test_scaled, type = "prob")[, "Yes"]  
roc.qda <- roc(actual, pred.qda, levels = c("No", "Yes"))  
plot(roc.qda, col = "green", lwd = 2, main = "ROC Curve: QDA")
```



```
cat("AUC QDA:", auc(roc.qda), "\n")
```

```
## AUC QDA: 0.6484842
```

Confusion Matrices

```
# Confusion Matrices
```

```
threshold <- 0.1
```

```
library(caret)
```

```
# Logistic models
```

```
simple_probs <- predict(simple, trainHeart, type = "response")
```

```
simple.pred <- factor(ifelse(simple_probs > threshold, "Yes", "No"), levels =  
c("No", "Yes"))
```

```
complex1_probs <- predict(complex1, trainHeart, type = "response")
```

```
complex1.pred <- factor(ifelse(complex1_probs > threshold, "Yes", "No"),  
levels = c("No", "Yes"))
```

```
complex2_probs <- predict(complex2, trainHeart, type = "response")
```

```
complex2.pred <- factor(ifelse(complex2_probs > threshold, "Yes", "No"),  
levels = c("No", "Yes"))
```

```
# LDA predictions
```

```
lda_probs <- predict(lda.fit, train_scaled, type = "prob")[, "Yes"]
```

```
lda.pred <- factor(ifelse(lda_probs > threshold, "Yes", "No"), levels =
```

```

c("No", "Yes"))

# QDA predictions
qda_probs <- predict(qda.fit, train_scaled, type = "prob")[, "Yes"]
qda.pred <- factor(ifelse(qda_probs > threshold, "Yes", "No"), levels =
c("No", "Yes"))

# KNN predictions
knn_probs <- predict(knn.fit, train_scaled, type = "prob")[, "Yes"]
knn.pred <- factor(ifelse(knn_probs > threshold, "Yes", "No"), levels =
c("No", "Yes"))

# Now create confusion matrices as before
cm.simple <- confusionMatrix(simple.pred, trainHeart$HeartDisease, positive =
"Yes")
cm.complex1 <- confusionMatrix(complex1.pred, trainHeart$HeartDisease,
positive = "Yes")
cm.complex2 <- confusionMatrix(complex2.pred, trainHeart$HeartDisease,
positive = "Yes")
cm.lda <- confusionMatrix(lda.pred, train_scaled$HeartDisease, positive =
"Yes")
cm.qda <- confusionMatrix(qda.pred, train_scaled$HeartDisease, positive =
"Yes")
cm.knn <- confusionMatrix(knn.pred, train_scaled$HeartDisease, positive =
"Yes")

# Print confusion matrices
cat("Confusion Matrix: Simple Logistic Regression\n")

## Confusion Matrix: Simple Logistic Regression

print(cm.simple)

## Confusion Matrix and Statistics
##
##           Reference
## Prediction      No    Yes
##           No 182405  5641
##           Yes  51533 16258
##
##           Accuracy : 0.7765
##           95% CI : (0.7749, 0.7781)
##           No Information Rate : 0.9144
##           P-Value [Acc > NIR] : 1
##
##           Kappa : 0.2678
##
##           McNemar's Test P-Value : <2e-16
##
##           Sensitivity : 0.74241

```

```

##             Specificity : 0.77972
##             Pos Pred Value : 0.23983
##             Neg Pred Value : 0.97000
##             Prevalence : 0.08560
##             Detection Rate : 0.06355
##             Detection Prevalence : 0.26498
##             Balanced Accuracy : 0.76106
##
##             'Positive' Class : Yes
##

cat("\nConfusion Matrix: Complex Logistic Regression 1\n")

##
## Confusion Matrix: Complex Logistic Regression 1

print(cm.complex1)

## Confusion Matrix and Statistics
##
##             Reference
## Prediction      No      Yes
##             No 179313   5211
##             Yes  54625  16688
##
##             Accuracy : 0.7661
##             95% CI : (0.7645, 0.7678)
##             No Information Rate : 0.9144
##             P-Value [Acc > NIR] : 1
##
##             Kappa : 0.2613
##
##             Mcnemar's Test P-Value : <2e-16
##
##             Sensitivity : 0.76204
##             Specificity : 0.76650
##             Pos Pred Value : 0.23401
##             Neg Pred Value : 0.97176
##             Prevalence : 0.08560
##             Detection Rate : 0.06523
##             Detection Prevalence : 0.27874
##             Balanced Accuracy : 0.76427
##
##             'Positive' Class : Yes
##

cat("\nConfusion Matrix: Complex Logistic Regression 2\n")

##
## Confusion Matrix: Complex Logistic Regression 2

```

```

print(cm.complex2)

## Confusion Matrix and Statistics
##
##           Reference
## Prediction    No    Yes
##           No 178461  5114
##           Yes 55477 16785
##
##           Accuracy : 0.7632
##           95% CI : (0.7615, 0.7648)
##           No Information Rate : 0.9144
##           P-Value [Acc > NIR] : 1
##
##           Kappa : 0.2592
##
##  Mcnemar's Test P-Value : <2e-16
##
##           Sensitivity : 0.76647
##           Specificity : 0.76286
##           Pos Pred Value : 0.23228
##           Neg Pred Value : 0.97214
##           Prevalence : 0.08560
##           Detection Rate : 0.06561
##           Detection Prevalence : 0.28245
##           Balanced Accuracy : 0.76466
##
##           'Positive' Class : Yes
##

cat("\nConfusion Matrix: LDA\n")

##
## Confusion Matrix: LDA

print(cm.lda)

## Confusion Matrix and Statistics
##
##           Reference
## Prediction    No    Yes
##           No 20641 1513
##           Yes 2759  670
##
##           Accuracy : 0.833
##           95% CI : (0.8284, 0.8376)
##           No Information Rate : 0.9147
##           P-Value [Acc > NIR] : 1
##
##           Kappa : 0.1502
##

```



```

## McNemar's Test P-Value : <2e-16
##
##          Sensitivity : 0.30692
##          Specificity : 0.88209
##          Pos Pred Value : 0.19539
##          Neg Pred Value : 0.93171
##          Prevalence : 0.08533
##          Detection Rate : 0.02619
##          Detection Prevalence : 0.13403
##          Balanced Accuracy : 0.59451
##
##          'Positive' Class : Yes
##

cat("\nConfusion Matrix: QDA\n")

##
## Confusion Matrix: QDA

print(cm.qda)

## Confusion Matrix and Statistics
##
##          Reference
## Prediction    No   Yes
##          No 19966 1494
##          Yes  3434   689
##
##          Accuracy : 0.8074
##          95% CI : (0.8025, 0.8122)
##          No Information Rate : 0.9147
##          P-Value [Acc > NIR] : 1
##
##          Kappa : 0.1204
##
## McNemar's Test P-Value : <2e-16
##
##          Sensitivity : 0.31562
##          Specificity : 0.85325
##          Pos Pred Value : 0.16711
##          Neg Pred Value : 0.93038
##          Prevalence : 0.08533
##          Detection Rate : 0.02693
##          Detection Prevalence : 0.16116
##          Balanced Accuracy : 0.58443
##
##          'Positive' Class : Yes
##

cat("\nConfusion Matrix: KNN\n")

```

```

##
## Confusion Matrix: KNN

print(cm.knn)

## Confusion Matrix and Statistics
##
##           Reference
## Prediction    No   Yes
##           No 15254   226
##           Yes  8146  1957
##
##           Accuracy : 0.6728
##           95% CI : (0.667, 0.6785)
##           No Information Rate : 0.9147
##           P-Value [Acc > NIR] : 1
##
##           Kappa : 0.2073
##
## Mcnemar's Test P-Value : <2e-16
##
##           Sensitivity : 0.89647
##           Specificity : 0.65188
##           Pos Pred Value : 0.19370
##           Neg Pred Value : 0.98540
##           Prevalence : 0.08533
##           Detection Rate : 0.07650
##           Detection Prevalence : 0.39491
##           Balanced Accuracy : 0.77418
##
##           'Positive' Class : Yes
##

model_names <- c("Simple", "Complex1", "Complex2", "LDA", "QDA", "KNN")

accuracies <- c(
  cm.simple$overall["Accuracy"],
  cm.complex1$overall["Accuracy"],
  cm.complex2$overall["Accuracy"],
  cm.lda$overall["Accuracy"],
  cm.qda$overall["Accuracy"],
  cm.knn$overall["Accuracy"]
)

sensitivities <- c(
  cm.simple$byClass["Sensitivity"],
  cm.complex1$byClass["Sensitivity"],
  cm.complex2$byClass["Sensitivity"],
  cm.lda$byClass["Sensitivity"],
  cm.qda$byClass["Sensitivity"],

```

```

    cm.knn$byClass["Sensitivity"]
  )

specificities <- c(
  cm.simple$byClass["Specificity"],
  cm.complex1$byClass["Specificity"],
  cm.complex2$byClass["Specificity"],
  cm.lda$byClass["Specificity"],
  cm.qda$byClass["Specificity"],
  cm.knn$byClass["Specificity"]
)

performance_table <- data.frame(
  Model = model_names,
  Accuracy = round(accuracies, 4),
  Sensitivity = round(sensitivities, 4),
  Specificity = round(specificities, 4)
)

print(performance_table)

##      Model Accuracy Sensitivity Specificity
## 1   Simple    0.7765      0.7424      0.7797
## 2 Complex1    0.7661      0.7620      0.7665
## 3 Complex2    0.7632      0.7665      0.7629
## 4      LDA    0.8330      0.3069      0.8821
## 5      QDA    0.8074      0.3156      0.8532
## 6      KNN    0.6728      0.8965      0.6519

cat("AUC LDA:", auc(roc.lda), "\n")

## AUC LDA: 0.6459434

cat("AUC QDA:", auc(roc.qda), "\n")

## AUC QDA: 0.6484842

cat("AUC KNN:", auc(roc.knn), "\n")

## AUC KNN: 0.5901466

```

Logistic Simple/Complex Model Metrics

```

# Logistic Simple/Complex Model Metrics with 0.3 threshold
# Function to compute metrics for a given model and dataset
get_metrics <- function(model, data, positive_class = "Yes", threshold = 0.3)
{
  # Predict probabilities
  probs <- predict(model, newdata = data, type = "response")

  # Convert to predicted class using custom cutoff

```

```

preds <- ifelse(probs >= threshold, positive_class, "No")

# Convert to factors with same levels as actuals
actual <- factor(data$HeartDisease, levels = c("No", positive_class))
preds <- factor(preds, levels = c("No", positive_class))

# Confusion matrix
cm <- confusionMatrix(preds, actual, positive = positive_class)

# Extract metrics
sens <- cm$byClass["Sensitivity"]
spec <- cm$byClass["Specificity"]
ppv <- cm$byClass["Pos Pred Value"]
npv <- cm$byClass["Neg Pred Value"]

# Prevalence
prevalence <- mean(actual == positive_class)

# AUROC
roc_obj <- roc(actual, probs, levels = c("No", positive_class))
auc_val <- auc(roc_obj)

return(data.frame(
  Sensitivity = sens,
  Specificity = spec,
  Prevalence = prevalence,
  PPV = ppv,
  NPV = npv,
  AUROC = auc_val
))
}

# Apply to each model
metrics_simple <- get_metrics(simple, trainHeart, threshold = 0.1)
metrics_complex1 <- get_metrics(complex1, trainHeart, threshold = 0.1)
metrics_complex2 <- get_metrics(complex2, trainHeart, threshold = 0.1)

# Combine results
results <- rbind(
  Simple = metrics_simple,
  Complex1 = metrics_complex1,
  Complex2 = metrics_complex2
)

print(results)

##           Sensitivity Specificity Prevalence          PPV          NPV          AUROC
## Simple           0.7424083    0.7797151 0.08559747 0.2398253 0.9700020 0.8409817

```

## Complex1	0.7620439	0.7664980	0.08559747	0.2340106	0.9717598	0.8419010
## Complex2	0.7664734	0.7628560	0.08559747	0.2322798	0.9721422	0.8434701